

Toward Real-Time Coordinated Jam-Absorption Driving: Fundamental-Diagram-Based Mechanism Analysis and Multi-Granularity Hierarchical Control

Abstract

Traffic oscillations on highways reduce traffic efficiency, increase rear-end collision risk, and worsen energy and environmental performance, making effective wave mitigation an important problem in intelligent transportation systems. Although jam-absorption driving (JAD) has shown promise in smoothing stop-and-go traffic, existing studies still lack a clear theoretical explanation of its efficiency-recovery mechanism and are largely restricted to single-vehicle, constant-speed control paradigms. To address these gaps, this study aims to clarify the performance mechanism of JAD from the perspective of the fundamental diagram and to develop a real-time multi-vehicle coordinated control framework for mixed traffic environments. Specifically, a hierarchical reinforcement learning framework is proposed to jointly select absorbing vehicles and optimize their longitudinal trajectories, with the Intelligent Driver Model adopted to formulate the control process and a joint objective balancing traffic delay and collision risk. Numerical results show that the proposed four-vehicle cooperative strategy significantly outperforms the uncontrolled case and conventional JAD, reducing TET, TIT, total delay, total stop time, and fuel consumption by 94.59%, 66.29%, 22.67%, 96.79%, and 15.28%, respectively, while improving driving comfort by 57.00%. The results further indicate that the proposed framework is robust under different CAV penetration rates and stochastic disturbances, providing both theoretical insight and methodological support for real-time cooperative JAD control.

1. Introduction

Traffic oscillations on highways may cause substantial capacity drops and reduced discharge rates, while also increasing the risk of rear-end collisions due to abrupt and frequent decelerations, as well as shorter headways, especially when drivers encounter the waves unexpectedly. Traffic oscillations result in capacity drops of 3% to 18% (Jerath et al., 2024; Knoop and Hoogendoorn, 2022), thereby increasing travel delays by 30% to 50% (Goldmann and Sieg, 2021; Fageda and Albalade, 2020). Rear-end crashes associated with traffic oscillations account for nearly 30% of highway accidents, primarily due to increased speed variation (Gaffney and Hovenden, 2023; Retallack and Ostendorf, 2019). Fuel consumption may increase by 10% to 40% (U.S. Department of Energy, n.d., 2024). Emissions of CO_2 and NO_x may also increase by 20% to 45% (Huang et al., 2018; He et al., 2024; Briani et al., 2024). Consequently, mitigating traffic oscillations is imperative for achieving efficient, safe, and sustainable mobility.

To address these issues, the concept of jam-absorption driving (JAD) has emerged as a promising active traffic control strategy. JAD employs a “slow-in, fast-out” driving logic, guiding vehicles to decelerate in advance before reaching a downstream traffic jam (Nishi et al., 2013). This maneuver transforms unstable stop-and-go traffic into smoother flow, thereby effectively absorbing the backward-propagating wave. He et al. (2016) improved the JAD strategy based on Newell’s car-following model. Simulation results showed that JAD can reduce total travel time (TTT) by approximately 0.67% and the number of vehicle stops by 44% to 47%. Zheng et al. (2020) and Wang et al. (2022) extended JAD to address collision risks. Simulation results indicate that JAD reduces TTT by 1.65% to 2.58%, time exposed time-to-collision (TET) by 35.76% to 44.22%, and time-integrated time-to-collision (TIT) by 34.41% to 41.19%.

To the best of our knowledge, existing JAD control systems generally rely on single-vehicle implementations. The exception is by Li et al. (2024), which proposed a rule-based real-time JAD strategy

for multiple waves. This strategy reduces travel time by 3% to 6%, improves inverse time-to-collision (iTTC) by 41% to 66%, and reduces fuel consumption by 7.5% to 11.6% in the test scenarios. Notably, this study assumes the deployment of at most one absorbing vehicle per congestion wave, rather than selecting multiple vehicles to accommodate traffic dynamics.

Hence, this study extends existing JAD research by selecting multiple vehicles to counteract traffic shock waves, thereby improving both traffic efficiency and safety. With the advent of CAV technology, vehicle trajectories can be controlled more precisely. Furthermore, we develop a hierarchical deep reinforcement learning algorithm for real-time JAD control with multi-vehicle coordination. The algorithm selects absorbing vehicles at the upper level and determines their trajectories at the lower level, with coordination mechanisms operating synergistically between the two levels. By orchestrating multiple vehicles simultaneously, the framework prevents local control failures and ensures that shockwaves are absorbed more rapidly and stably. Moreover, given that JAD involves transitions among multiple regimes of the traffic fundamental diagram (FD), we link JAD to FD attributes rather than limiting the analysis to microscopic traffic simulations. This framework may provide theoretical insights into JAD performance on a more rigorous basis.

The remainder of this paper is organized as follows. [Section 2](#) reviews the relevant literature on jam-absorption driving (JAD). [Section 3](#) presents the theoretical analysis of JAD and provides insights from the fundamental diagram. [Section 4](#) develops the multi-vehicle coordinated JAD control framework, including the problem formulation, modeling assumptions, control strategy, and solution algorithm. [Section 5](#) evaluates the performance of the proposed method through numerical experiments under different CAV penetration rates, heterogeneous traffic conditions, and stochastic driving disturbances. Finally, [Section 6](#) concludes the paper and discusses future research directions.

2. Literature review

2.1. Concept and Mechanism of JAD

The core philosophy of JAD diverges from standard reactive driving. Instead of maintaining a tight gap with the preceding vehicle, the absorbing vehicle acts as a "dampener" within the traffic stream. That is, the absorbing vehicle adopts counter-intuitive "slow-in, fast-out" logic. As elucidated by [Nishi et al. \(2013, 2020\)](#), when the absorbing vehicle detects a downstream jam, it voluntarily initiates deceleration well before reaching the congestion tail. This early deceleration creates a substantial spatial and temporal gap between the absorber and the preceding platoon. When the backward-propagating kinematic wave reaches the absorbing vehicle, it is effectively "digested" within the spatial and temporal gap. Consequently, the absorbing vehicle can pass through the congested area at a relatively constant low speed without coming to a complete stop, thereby preventing the shockwave from propagating further upstream to the following vehicles.

Concurrently, the deceleration phase of the absorbing vehicle generates upstream perturbations that may evolve into secondary jams due to flow instability. [Taniguchi et al. \(2015\)](#) analyzed this phenomenon using microscopic car-following models, determining specific parameter regions for the absorbing velocity and duration that effectively suppress these secondary waves. Furthermore, [Nishi \(2020\)](#) established theoretical conditions based on linear string stability to guarantee the restriction of secondary jams. These studies validate that JAD can robustly improve traffic flow without inducing upstream congestion, even in complex scenarios with bottlenecks or lane inflows.

Leveraging Newell’s car-following theory (Newell, 2002), He et al. (2017) improved the JAD strategy by optimizing the control spacetime range and absorbing speed, and employed a search method to identify the optimal absorbing vehicle. By aligning the absorbing vehicle’s trajectory with the wave propagation path, JAD allows traffic density to reset from a congested state to a capacity or free-flow state (Taniguchi et al., 2015). This process may smooth the driving profile of the absorbing vehicle and enable the following platoon to maintain a stable speed. Simulation shows that absorbing vehicle trajectories can not only dissipate shockwaves but also reduce total stops by 44% to 47% and TTT by 0.67%.

JAD also contributes to safety improvement. Zheng et al. (2020) found that JAD strategies could be optimized to minimize rear-end collision risks. By incorporating wavelet transforms to refine control intervals and accommodate vehicle speeds, their model significantly avoids abrupt braking by trailing vehicles. This approach reduces TET by 21%, TIT by 27%, and TTT by 1.27% in the simulation.

In recent years, the integration of Connected and Autonomous Vehicle (CAV) technologies has pushed JAD research into more complex scenarios. Wang et al. (2022) incorporated CAV market-penetration rate (MPR) from 10% to 100% into the JAD strategy and constructed a delay-based optimization objective function. Simulation results demonstrated that TTT, TET, and TIT were significantly reduced by 2.11%–5.58%, 14.63%–44.22%, and 12.8%–41.19%, respectively. Addressing the limitation of single-wave scenarios, Li et al. (2024) developed a system capable of handling multiple mobile blockages. This work introduced the concept of real-time control into the JAD framework for the first time, utilizing estimation algorithms to dynamically adjust the speed of absorbing vehicles. Through experiments across simulated scenarios with varying CAV MPR (1% to 50%), their optimal strategy achieves 3% to 6% travel time gains, 30% to 50% driving comfort improvement, 7.5% to 11.6% fuel saving improvement, 41% to 66% iTTC reduction, and 6.5% to 9% decrease in jam proportion. Similarly, Shen et al. (2025) employed a stochastic car-following model to confirm the elimination of single and double motion waves, achieving reductions in TET by 15.45%, TIT by 15.77%, speed variability by 16.22%, and energy consumption by 6.51%, while cutting CO_2 , NO_x , and PM emissions by 25.05%, 30.30%, and 31.53%, respectively. Furthermore, Liu et al. (2025) extended the consensus to mixed traffic flows, demonstrating that longitudinal control of a single CAV as an absorber vehicle can mitigate oscillations even at low automation penetration rates. In their simulation experiments, this approach optimized average vehicle speed by 1.6%, TET by 70%, fuel consumption by 9%, and reduced vehicle stop times (VST) by 50%. Collectively, these studies suggest a broad consensus that JAD is a versatile and effective tool for stabilizing traffic flow, improving safety, and reducing environmental impacts across diverse traffic scenarios.

2.2. Theoretical Perspectives and Emerging Insights on JAD Performance

The initial theoretical frameworks of JAD, primarily established by Nishi et al. (2013), were grounded in fundamental-diagram (FD) analysis. While these early studies mathematically proved that a “slow-in, fast-out” maneuver could successfully dissipate the kinematic structure of a jam and smooth traffic flow, they did not explicitly demonstrate that this smoothing effect translates into improved macroscopic traffic efficiency (e.g., reduced total travel time or delay). In contrast, subsequent research utilizing advanced optimization algorithms has consistently reported positive impacts on efficiency. Studies employing trajectory optimization (He et al., 2017; Wang et al., 2022) and optimal control strategies (Zheng et al., 2020; Li et al. 2024; Liu et al., 2025) have empirically or numerically shown that the implementation of JAD reduces total travel time. These findings are largely phenomenological, with specific optimization outcomes depending primarily on scenario-specific parameter settings, such as initial traffic conditions and destination locations. Most existing studies treat efficiency gains as stable by-products without rigorously decoupling oscillation-absorption mechanisms from capacity-recovery mechanisms. Consequently, the

theoretical link between microscopic trajectory smoothing and macro-level improvements in flow efficiency remains a “black box,” lacking a comprehensive physical explanation within the current knowledge framework.

Regarding control strategies for JAD, prevailing paradigms typically operate under a "single-vehicle-single-wave" (He et al., 2017; Zheng et al., 2020; Wang et al., 2022) or "single-vehicle-multi-wave" (Li et al., 2024) configuration. In these frameworks, the control logic is predominantly predicated on constant-speed cruising strategies. While this uniform-velocity approach simplifies theoretical analysis, it inherently imposes a ceiling on the system's performance by failing to exploit the dynamic acceleration and deceleration capabilities of CAVs. Furthermore, achieving robust real-time control remains a formidable challenge. Although recent studies such as Li et al. (2024) have developed systems for real-time jam estimation, the dual challenge of dynamically selecting the optimal absorbing vehicle and precisely controlling its trajectory in a rapidly evolving traffic stream remains an unresolved bottleneck in current methodologies.

To break these boundaries, emerging studies offer critical insights into potential solutions. Regarding the performance limit, Suzuki et al. (2025) explored the concept of string-stable support vehicles, providing empirical evidence that multi-vehicle coordination can yield significantly higher stability and optimization effects compared to isolated agents. Regarding the challenge of real-time execution, Kreidieh et al. (2018) demonstrated the efficacy of deep reinforcement learning (DRL) in dissipating stop-and-go waves. Their work suggests that learning-based algorithms can effectively handle the high-dimensional state spaces of traffic flows, thereby offering a viable pathway for implementing real-time, responsive control strategies that are difficult to model using traditional heuristics.

Building upon these insights, this study constructs a real-time multi-vehicle coordinated variable-speed JAD framework based on Hierarchical Reinforcement Learning (HRL). Unlike previous approaches that treat selection and control as separate or static tasks, our framework establishes a complete decision-making chain, from the selection of absorbing vehicles to coordination-scheme formulation and, finally, to lower-level trajectory control. By integrating this hierarchical decision process, we aim to transcend the limitations of constant-speed strategies and single-vehicle paradigms, thereby exploring the optimal performance boundary of JAD in complex, high-density oscillation scenarios.

2.3 Summary of Research Gaps and Study Positioning

The literature on jam-absorption driving (JAD) has established its efficacy in mitigating traffic oscillations and enhancing stability; however, two major knowledge gaps remain.

(1) Lack of a theoretical explanation for the efficiency retrieval mechanism. There is a distinct disconnect between the phenomenological observation of efficiency gains and the physical explanation for their origin. Although previous studies generally report that JAD yields efficiency gains, rigorous analysis and interpretation remain lacking.

(2) Rigid constraints of control paradigms limit performance boundaries. Prevailing control strategies are predominantly confined to single-vehicle, constant-speed configurations. This reliance on rigid heuristics imposes a performance ceiling, rendering the system incapable of handling high-density or multi-wave oscillations. Furthermore, a unified framework for real-time execution is still lacking; specifically, the challenge of dynamically linking vehicle selection and trajectory control remains an unresolved bottleneck in complex traffic environments.

This study addresses these two gaps by (i) providing a fundamental-diagram-based explanation for why JAD can improve efficiency and (ii) proposing a real-time multi-vehicle coordinated control framework based on hierarchical reinforcement learning (HRL). By establishing a closed-loop link from decision-making to control and moving beyond the single-vehicle constant-speed paradigm, this research endeavors to explore the upper boundaries of JAD performance in mixed CAV environments.

3. JAD theory and Insights from fundamental diagram

3.1 JAD strategy

In a single-lane traffic stream, speed disturbances caused by the deceleration of the first vehicle propagate upstream and may evolve into traffic oscillations or congestion if left uncontrolled. JAD applies a two-phase “slow-in-fast-out” control strategy: during the slow-in phase, the absorbing vehicle decelerates to a prescribed speed and spacing to create sufficient buffer space for absorbing incoming oscillation waves; during the fast-out phase, it accelerates back to its initial speed after the disturbance is absorbed, thereby restoring steady traffic conditions without reducing overall efficiency.

For control implementation, when the first vehicle begins decelerating at position S_0 , the position of the absorbing vehicle at the same moment is denoted by S_i (Fig. 1). The control starting time t^s is defined as the instant at which the absorbing vehicle detects the arrival of the leading edge of the oscillation wave and initiates the slow-in maneuver. The absorption process terminates when the absorbing vehicle reaches a steady-state equilibrium, at which point its speed, acceleration, and inter-vehicle spacing return to equilibrium values and are no longer affected by upstream disturbances.

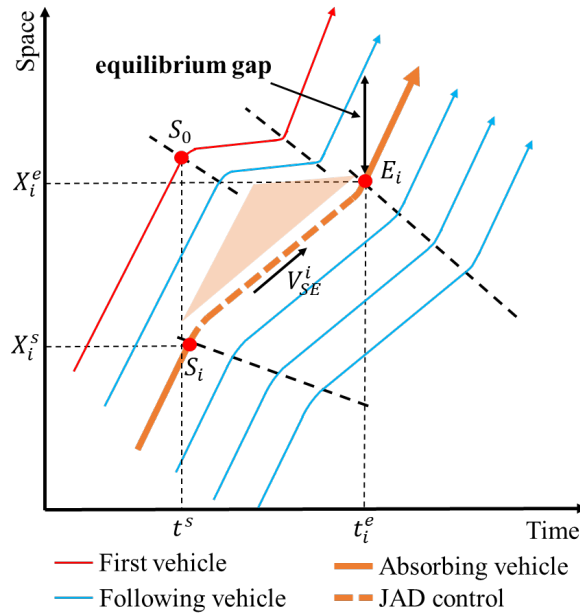


Fig. 1. JAD Strategy.

When the absorption process is completed, the position of the absorbing vehicle is defined as E_i , which corresponds to the termination point of the oscillation wave. At this moment, the vehicle speed satisfies $v(i, t) > v_{SE}^i$, indicating that it exceeds the control speed maintained during the absorption phase. Therefore, the vehicle is considered to have returned to a free-flow state. After the dissipation of traffic oscillations, the system is required to satisfy **steady-state traffic flow** conditions, given by

$$a(i, t) = 0 \quad (0)$$

$$\Delta v = 0 \quad (0)$$

$$s = s^e(i, v) \quad (0)$$

During the “Slow in–Fast out” jam-absorption driving control process, the absorbing vehicle travels at a constant absorption speed v_{SE}^i between the two spatiotemporal points S_i and E_i . This speed is calculated as follows:

$$v_{SE}^i = \frac{X_i^e - X_i^s}{t_i^e - t^s} \quad ()$$

where X_i^s and X_i^e denote the spatial positions of the absorbing vehicle at the beginning and the end of the control process, respectively, while t^s and t_i^e represent the corresponding start and end times. By maintaining the absorbing vehicle at a constant speed v_{SE}^i , a stable absorption region can be formed in the space-time diagram, thereby effectively attenuating the oscillation waves propagating upstream.

3.2 Efficiency Optimization Dispute

In previous studies on jam-absorption driving strategies, researchers have primarily focused on improving traffic efficiency, typically by adjusting the control strategy of the absorbing vehicle to minimize total delay or TTT. Assuming that the destination in the scenario is X_{end} , the total traffic delay can be calculated as follows:

$$D = \sum_{i=0}^N \frac{X_{end}}{v_0} - TTT_i \quad ()$$

where D is total delay, X_{end} is hypothetical road destination, and TTT_i is total travel time of vehicle i . According to macroscopic traffic flow theory, after experiencing congestion, the traffic state will return to the capacity state (Q_M, K_M, V_M) .

Therefore, once the trajectory of the first vehicle is determined, the final total delay does not depend on the trajectories of the absorbing vehicles in the scenario:

$$D = \sum_{i=0}^N \frac{X_{end}}{v_0} - TTT_i = \sum_{i=0}^N \frac{X_{end}}{v_0} - (TTT_0 + \frac{N}{Q_M}) \quad ()$$

This expression indicates that, from a macroscopic perspective, the final total delay is jointly determined by the trajectory of the first vehicle and the road-capacity characteristics, and is independent of the specific trajectories of the absorbing vehicles (Fig. 2).

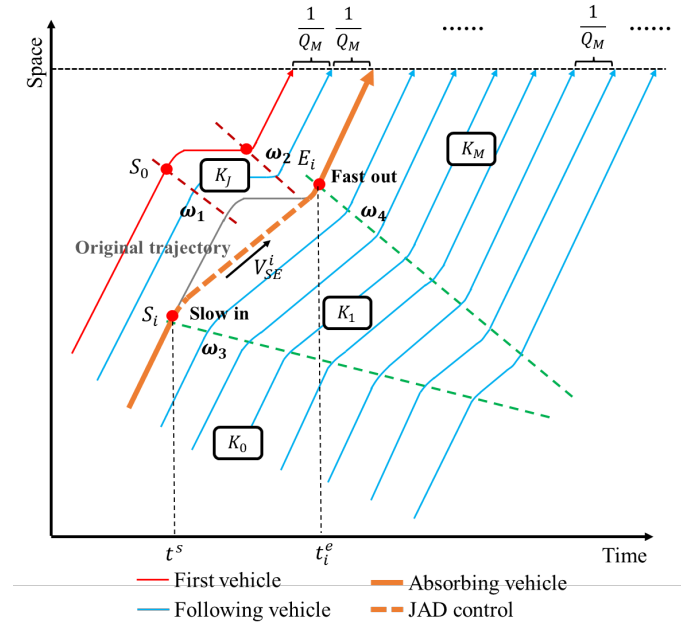


Fig. 2. Principle of efficiency optimization dispute.

The JAD strategy commonly involves transitions among traffic states, such as jam, free-flow, capacity, and congested states. Therefore, JAD can be interpreted within the framework of the fundamental diagram, which may impose a theoretical boundary on changes in traffic efficiency. According to the initial traffic state before the

shockwave occurs, three cases are discussed, with the capacity drop at critical density considered in Case 3.

Case 1: initial traffic state being non-congested. Fig. 3 compares the effect of shockwave on vehicle trajectory without JAD, with predictable JAD, and with practical JAD. Fig. 3(a) shows vehicle trajectory without JAD. Fig. 3(b) illustrates a predictable JAD scenario in which traffic congestion can be anticipated in advance. In this scenario, the optimal strategy involves controlling the vehicle immediately following the jam-inducing vehicle to maintain a speed corresponding to the capacity state (state C). Consequently, the traffic flow transitions directly to the capacity state, thereby avoiding the adverse effects associated with entering the congested state. Fig. 3(c) presents a practical JAD scenario in which congestion cannot be predicted in advance. In this case, the traffic state can similarly be switched directly to capacity by controlling the subsequent absorbing vehicle to maintain a speed corresponding to capacity. This approach introduces an inherent trade-off. If the absorbing vehicle is positioned too far downstream, a lower absorbing speed may cause subsequent vehicles to transition into the congested state. Conversely, if the vehicle is located too far upstream, downstream traffic will still be exposed to the jam, thereby increasing the risk of accidents. In this case, irrespective of whether the JAD strategy is deployed, the traffic stream eventually settles into a car-following regime behind the jam-inducing vehicle at the capacity state. Therefore, the strategy does not provide improvements in efficiency-related metrics (e.g., TTT and delay).

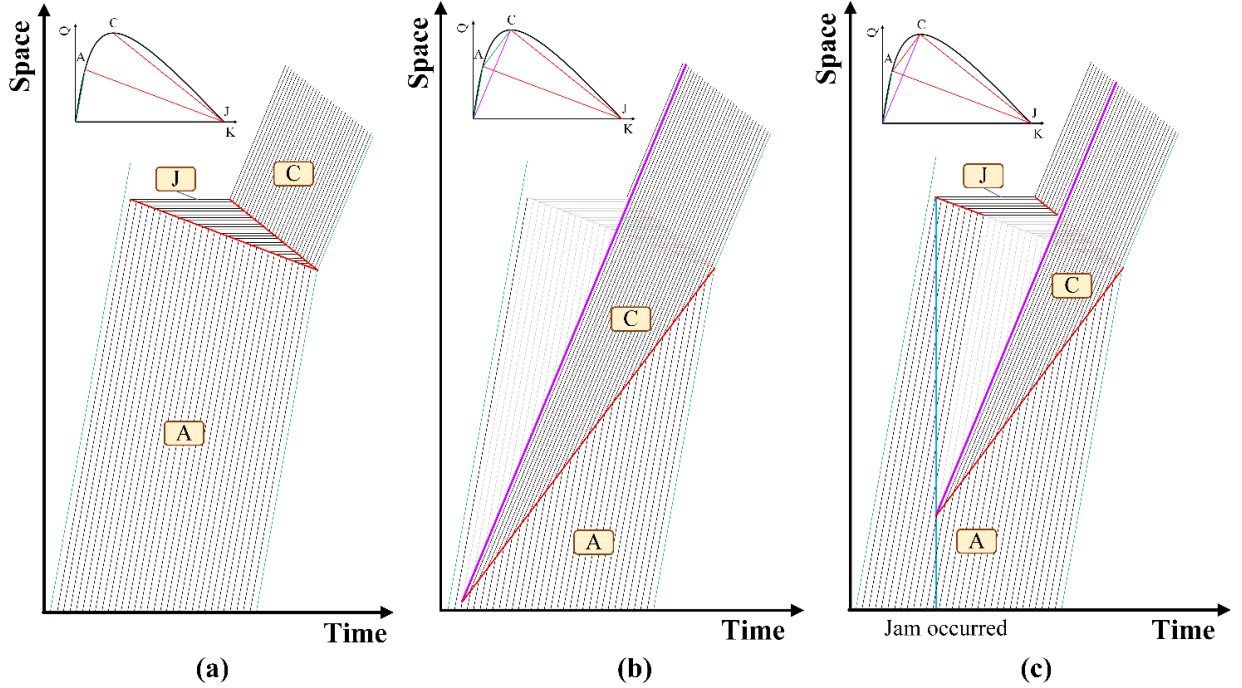


Fig. 3. Equilibrium scenario.

Case 2: initial traffic state being congested. Fig. 4(a) shows vehicle trajectory without JAD. Fig. 4(b) illustrates the predictable JAD case. In this scenario, since the absorbing speed cannot attain the capacity speed, the vehicle is constrained to travel at a sufficiently upstream location in a state (e.g., State B) as close as possible to the initial speed. Fig. 4(c) is practical JAD. Similarly, an absorbing vehicle with an absorbing speed as close as possible to the initial speed is selected for control, thereby minimizing the risks within the scenario to the greatest extent. In this case, the traffic wave induced by the JAD strategy cannot be fully dissipated. Consequently, the primary contribution of JAD lies in enhancing safety and driving comfort, whereas the strategy yields no efficiency gains.

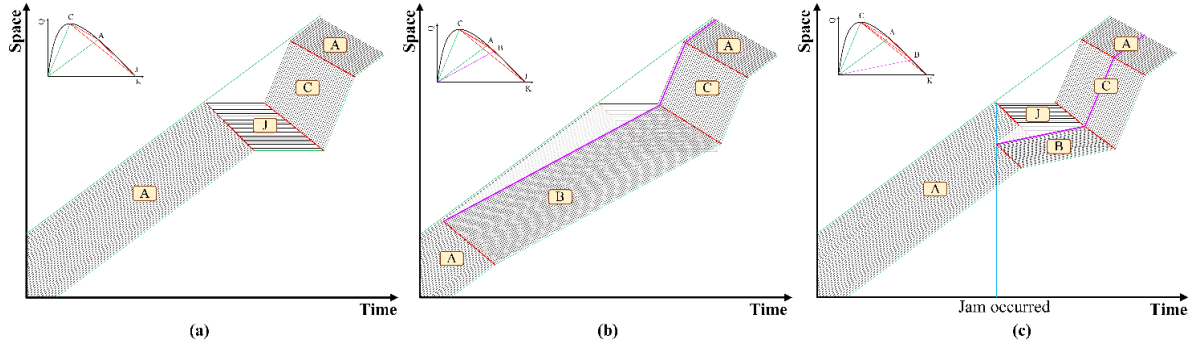


Fig. 4. Congestion scenario.

Case 3: effect of capacity drop. Fig.5(a) shows vehicle trajectory without JAD. Fig.5(b) illustrates the predictable JAD case. By controlling the vehicle immediately following jam-inducing vehicle to travel at capacity speed in advance, the upstream traffic avoids experiencing the congested state and transitions directly into a state of higher capacity (state C). Regarding the realistic JAD in Fig.5(c), although an absorbing vehicle can still be identified and controlled to travel at capacity speed, downstream vehicles inevitably experience the congested state. Consequently, the final traffic state remains constrained to a lower capacity (state C'). In this case, the predictable JAD strategy is the only approach that yields efficiency gains.

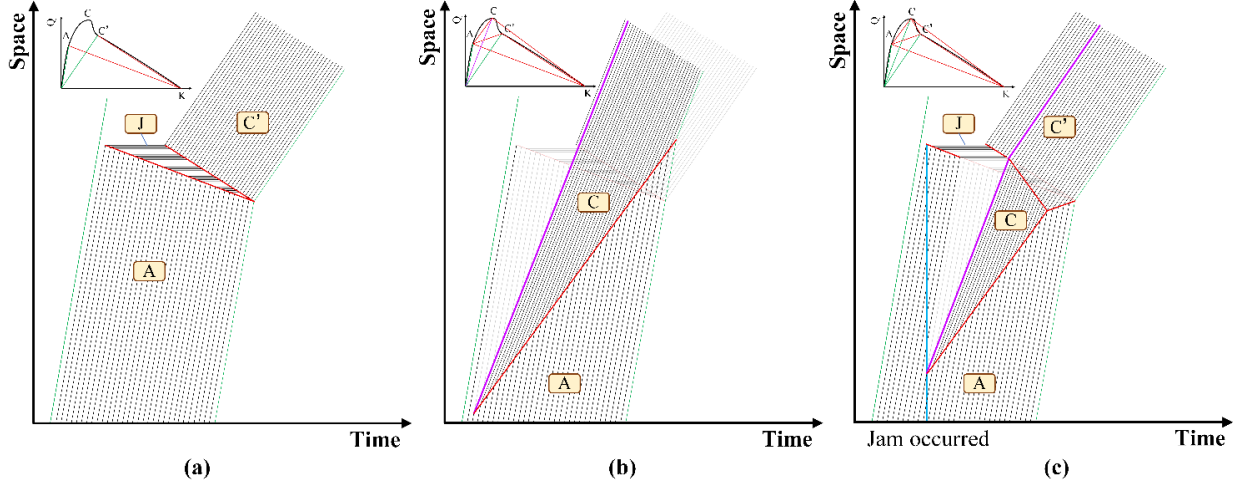


Fig. 5. Capacity drop scenario.

In actual microscopic scenarios, traffic flow does not shift instantaneously from congestion to capacity, but rather undergoes a gradual transition. Taking a simulation with 100 vehicles as an example, the process by which traffic flow evolves from congestion to capacity can be influenced by the JAD strategy, thereby reducing traffic delay during this transition (Fig. 6). Therefore, when the destination X_{end} is close to the congestion location, the traffic flow has not yet switched to the capacity state, and the traffic delay can be optimized.

$$D = \sum_{i=0}^N \frac{X_{end}}{v^0} - TTT_i < \sum_{i=0}^N \frac{X_{end}}{v^0} - (TTT_0 + \frac{N}{Q_M}) \quad ()$$

The farther the destination is from the oscillation point, the smaller the delay reduction becomes, and it eventually approaches 0 when $x \geq X$.

$$D'_{100}(x = 0.5) > D'_{100}(x = 1.5) > D'_{100}(x = 2.5) > D'_{100}(x \geq X) = 0 \quad ()$$

where $D'_i(x)$ denotes the delay reduction of vehicle i when the destination is x , and X denotes the road location at which the traffic flow reaches the capacity state. This demonstrates that the delay optimization induced by JAD is only effective during the local phase in which traffic flow has not yet fully recovered to equilibrium state. Once the system reaches the capacity, its impact on overall traffic efficiency completely vanishes. Macroscopic traffic flow theory determines the final state, whereas microscopic traffic flow theory provides the optimization targets during

the traffic-state transition process.

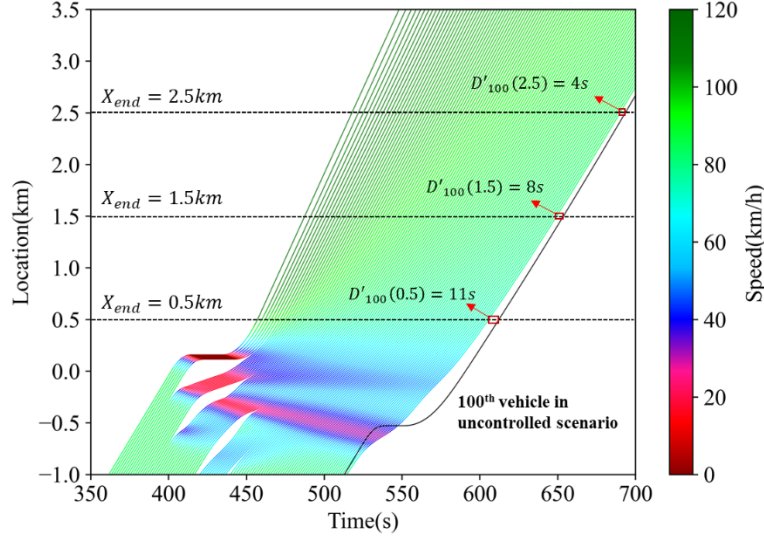


Fig. 6. Effect of efficiency improvement on evaluation points.

4. JAD control with multiple absorbing vehicles

4.1 Problem illustration

To isolate the effect of the proposed multi-vehicle coordinated control strategy from external disturbances, all vehicles are assumed to travel on a closed single-lane ring road. This closed-loop setting ensures that the formation and dissipation of traffic oscillations are governed solely by the internal traffic dynamics, while also facilitating control implementation and spatiotemporal observation of wave evolution. In this scenario, vehicle states such as position, speed, acceleration, and desired time headway are obtained through V2X communication, enabling accurate perception and cooperative trajectory control. The resulting high-resolution spatiotemporal information provides an appropriate testbed for multi-agent coordinated control. To quantify the spatial propagation characteristics of traffic oscillations, as shown in Fig. 7, the vehicle platoon on the ring road is divided into an upstream region, S_u , and a downstream region, S_d . The upstream region S_u was defined as the road segment immediately behind the first vehicle, opposite to the direction of oscillation propagation. Vehicles in this region are affected first by variations in the first vehicle's speed, making this region the primary observation window for the backward propagation of traffic shock waves. In contrast, the downstream region S_d is located ahead of the first vehicle. This region reflects the residual effects of the disturbance after it has propagated once through the entire platoon and is mainly used to evaluate the effectiveness of the control strategy in suppressing the cyclic amplification of oscillations and maintaining global system stability.

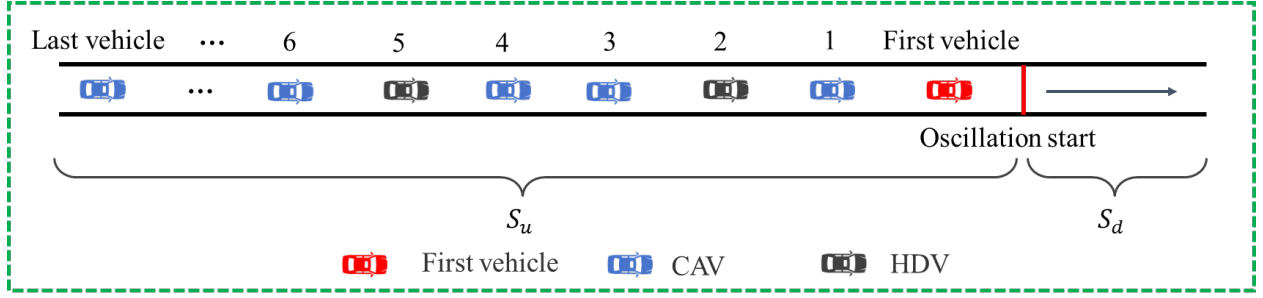


Fig. 7. Single-lane scenario.

A group of absorbing vehicles is selected from the target region and coordinated during the driving process. The target region is defined as starting at the location of first vehicle and extending upstream along S_u . All vehicles within this region are included in the coordinated JAD control scheme. An algorithm is employed to automatically select χ vehicles, indexed as $J_1, J_2, \dots, J_\chi$, to serve as absorbing vehicles (Fig. 8). The selected absorbing vehicles execute cooperative longitudinal speed control strategies at each time interval according to their respective positions and dynamic characteristics.

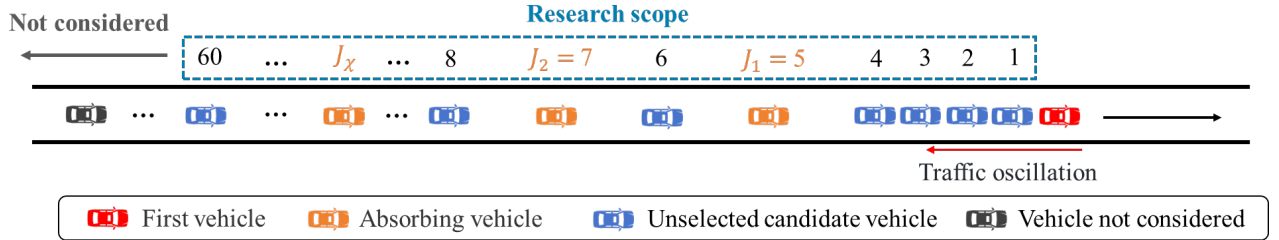


Fig. 8. Concept of multi-vehicle coordinated control.

Vehicle selection is performed at the start of the multi-vehicle coordinated JAD process and is maintained throughout the control process for the selected vehicles. This design may help reduce disturbances to vehicle operation and lower the computational cost of the proposed method. Vehicle selection is implemented referring to the primary traffic state of vehicle speed and locations for all vehicles in the research scope. The proposed JAD strategy tunes vehicle acceleration to coordinate their selected vehicles to attenuate the chock wave passed from the first vehicle.

4.2 Basic assumptions

For the sake of tractability, the following assumptions are adopted in the scenario model:

- 1) CAVs are assumed to possess accurate sensing and communication capabilities and to execute the driving commands generated by the control system with precision.
- 2) CAVs are assumed to travel only in the longitudinal direction; lane-changing behavior and lateral disturbances are excluded from consideration.
- 3) All CAVs are assumed to follow the same set of parameters in the vehicle dynamics model.
- 4) The acceleration and deceleration processes of CAVs are jointly constrained by vehicle-dynamics limits and safety requirements.
- 5) Road geometry and speed limits are assumed to remain unchanged throughout the simulation, and the effects of external boundary conditions, such as ramps and signal control, on mainline traffic flow are not considered.
- 6) In the mixed-traffic environment, HDVs are assumed to operate autonomously according to a predefined car-following rule and not to actively respond to the control strategy. CAVs follow the same underlying car-following

model, with additional acceleration or speed commands applied as external control inputs.

7) The set of CAVs and the market penetration rate are fixed within each experiment.

8) To ensure consistency in evaluation, the efficiency metric is measured at a designated downstream evaluation point.

These assumptions simplify system complexity while preserving the essential characteristics of real traffic-flow dynamics, thereby providing a sound basis for the subsequent validation of the control strategy.

4.3 Multi-vehicle Coordinated JAD Strategy

4.3.1 IDM car-following model

This study utilizes the Intelligent Driver Model (IDM), a microscopic car-following model proposed by Treiber et al. (2000), to simulate longitudinal traffic dynamics.

The acceleration equation of the IDM is given as follows:

$$a_{IDM_{i,t}} = a_{max} \left[1 - \left(\frac{v_{i,t}}{v_d} \right)^\delta - \left(\frac{s_{d_{i,t}}}{s_{i,t}} \right)^2 \right] \quad ()$$

where $a_{IDM}(i, t)$ denotes the longitudinal acceleration of vehicle i at time t ; a_{max} is maximum comfortable acceleration of the vehicle; $v(i, t)$ is the actual speed of vehicle i at time t ; v_d is desired target speed of the driver; δ is free-flow acceleration exponent, which characterizes the acceleration behavior of the vehicle under free-flow conditions; $s(i, t)$ is actual spacing between vehicle i and its preceding vehicle; and $s_d(i, t)$ is the desired safe spacing maintained by the driver.

The actual spacing between vehicles is defined as:

$$s_{i,t} = X_{i-1,t} - X_{i,t} - L \quad ()$$

where $X(i, t)$ is position of vehicle i at time t , and L is length of the preceding vehicle. This expression reflects the net spacing obtained by subtracting the length of the leading vehicle from the positional difference between the two vehicles.

The desired safe spacing is defined as:

$$s_{d_{i,t}} = s_0 + \max \left(T_d v_{i,t} + \frac{v_{i,t} \Delta v_{i,t}}{2\sqrt{a_{max} b_{max}}}, 0 \right) \quad ()$$

where s_0 is minimum safe spacing under standstill conditions; T_d is the desired time headway of the driver; $\Delta v(i, t) = v(i, t) - v(i-1, t)$ is relative speed with respect to the preceding vehicle; and b_{max} is maximum comfortable deceleration. This formulation jointly accounts for safety distance maintenance and speed adaptation mechanisms.

4.3.2. Equilibrium state

By incorporating IDM, the steady-state condition can be expressed as:

$$0 = a_{max} \left[1 - \left(\frac{v_{i,t}}{v_d} \right)^\delta - \left(\frac{s_{d_{i,t}}}{s_{e(i,v)}} \right)^2 \right] \quad ()$$

The steady-state spacing function is given by:

$$s_{e(i,v)} = s_0 + \frac{v_{i,t} T_d}{\sqrt{1 - \left(\frac{v_{i,t}}{v_d} \right)^\delta}} \quad ()$$

This function indicates that, under equilibrium conditions, a nonlinear relationship exists between vehicle

spacing and speed, and that steady-state spacing increases with vehicle speed. When the absorbing vehicle reaches this equilibrium spacing, it signifies that the energy of the oscillation wave has been fully absorbed and that traffic flow has been restored to a stable state.

4.3.3 Decision variables and constraints

Multi-vehicle coordinated JAD selects a set of absorbing vehicles, $J = [J_1, J_2, \dots, J_k, \dots, J_\chi]$, at the beginning of the control process. Notation J_k denotes the k -th absorbing vehicle, and χ represents the total number of absorbing vehicles. JAD adjusts the acceleration $a_J(J_k, t)$ of selected vehicle $k \in [1, 2, \dots, \chi]$ in real time for $t \in [1, 2, \dots, T]$, where T denotes the total number of control steps. To ensure real-time system responsiveness, the control interval Δt for acceleration updates is set to 1 s. Decision variable $a_J(J_k, t)$ is bounded with

$$-b_{\max} \leq a_J(J_k, t) \leq a_{IDM}(J_k, t) \quad ()$$

where $a_{IDM}(J_k, t)$ is the theoretical maximum acceleration calculated by IDM. This constraint ensures that, during congestion-absorption tasks, the acceleration of the absorbing vehicles remains within a physically achievable range, thereby guaranteeing driving smoothness and comfort.

In terms of control timing, the activation time of an absorbing vehicle is denoted by t^s , which represents the start of control, while the control end time is denoted by $t_i^{e'}$. In practical computations, to expand the solution space and enhance the adaptability of the control strategy, this study defines the end time of absorption control as the original end time t_i^e plus a fixed duration ζ , i.e.,

$$t_i^{e'} = t_i^e + \zeta \quad ()$$

where $\zeta = 10$ s indicates that, after oscillation absorption is completed, control is maintained for an additional period to ensure a smooth transition of traffic flow. The absorbing vehicles execute speed and acceleration adjustments within the interval $[t^s, t_i^{e'}]$, and their trajectory variations can generate a coordinated absorption effect with adjacent vehicles, thereby achieving spatiotemporal cooperation among multiple vehicles and effectively reducing the energy and propagation speed of oscillation waves (Fig. 9).

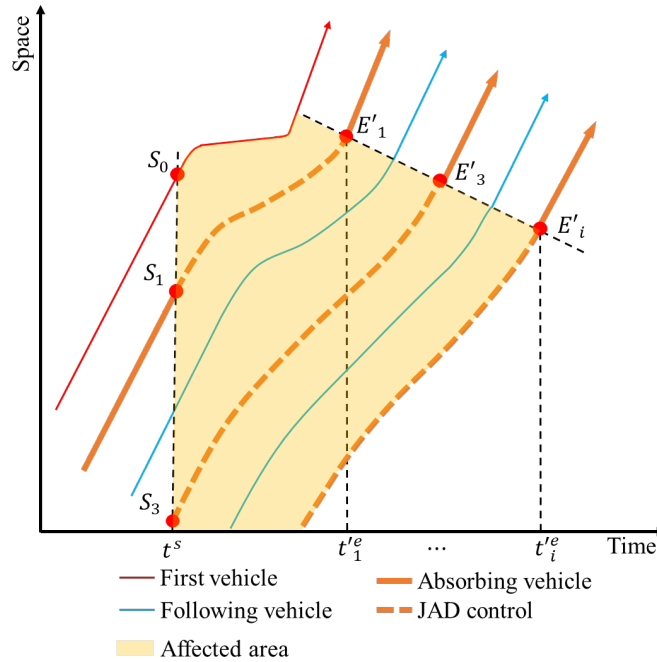


Fig. 9. Multi-vehicle coordination control diagram.

4.3.4 Control objective

Multi-vehicle coordinated JAD should be designed to jointly optimize traffic efficiency and safety. Otherwise, vehicles may exhibit frequent accelerations and decelerations in pursuit of efficiency, which may lead to excessively small inter-vehicle distances or abrupt acceleration changes, thereby increasing the potential collision risk; alternatively, they may become overly conservative for safety, resulting in slower traffic recovery. The objective function of multi-vehicle coordinated JAD control is given by

$$\min Z = \sum_{t=0}^T [Z_1(t) + \omega Z_2(t)] \quad ()$$

which dynamically balances traffic efficiency and safety. Here, $Z_1(t)$ represents traffic delay at time t , $Z_2(t)$ represents the safety evaluation at time t , and ω denotes the safety-weighting coefficient. We set $\omega = 1$, indicating that efficiency and safety are assigned equal importance in JAD control. Traffic delay $Z_1(t)$ reflects the overall operational efficiency of traffic flow and is given by

$$Z_1(t) = \sum_{i=1}^N \frac{d(i,t)}{D_{\max}} \quad ()$$

where $d(i,t)$ represents the relative delay of vehicle i , and $D_{\max}$ is the maximum total delay in the uncontrolled scenario. The quantity $d(i,t)$ is calculated as

$$d_{i,t} = 1 - \frac{X(i,t) - X(i,t-1)}{v_0} \quad ()$$

where $X(i,t)$ is the position of vehicle i at time t , and v_0 free-flow speed. Variable $d(i,t)$ measures the impact of traffic oscillations on overall traffic efficiency with the ratio of vehicle displacement to free-flow speed. To prevent over-optimization from causing excessively slow traffic recovery, the evaluation endpoint is set 2 km downstream of the oscillation starting point.

The safety indicator $Z_2(t)$ measures the potential collision risk among vehicles during the JAD control process and is defined as follows:

$$Z_2(t) = \sum_{i=1}^N \frac{TIT(i,t)\Delta t}{TIT_{\max}} \quad ()$$

where $TIT(i,t)$ denotes the Time-Integrated Time-to-Collision (TIT) of vehicle i , reflecting potential collision risk throughout the control period. Its computation is given by:

$$TIT_{i,t} = \begin{cases} \frac{1}{T} \sum (TTC_{i,t-1} - TTC_0), & \text{if } TTC_{i,t} < TTC_0 \\ 0, & \text{otherwise} \end{cases} \quad ()$$

where $TTC(i,t)$ is Time-to-Collision between vehicle i and its preceding vehicle, i.e., the time needed for vehicle i to collide with the leading vehicle if both maintain current speeds, defined as:

$$TTC_{i,t} = \begin{cases} \frac{s(i,t)}{v_{i,t} - v_{i-1,t}}, & v_{i,t} > v_{i-1,t} \\ \infty, & v_{i,t} \leq v_{i-1,t} \end{cases} \quad ()$$

where $s(i,t)$ is inter-vehicle spacing. When $TTC(i,t)$ is less than the threshold $TTC_0 = 3s$, a potential safety risk is considered to exist between vehicles, and the system increases $Z_2(t)$ to guide the control algorithm toward safer behavior. If $TTC(i,t)$ exceeds the threshold, the driving state is considered safe, and no further adjustment is required. By integrating TIT over time, the overall safety risk of the vehicle platoon during the control period can be comprehensively quantified, thereby ensuring that oscillation absorption is performed within safe boundaries.

4. Solution for JAD control of multiple vehicles

4.1 Algorithm structure

4.1.1 Framework

The Multi-Granularity Hierarchical Multi-Agent Coordination Policy Optimization (MG-HiMaPO) framework decomposes the overall decision-making process into a higher-level strategic layer and a lower-level control layer, thereby enabling efficient coordination between global decision-making and local vehicle control (Fig. 10).

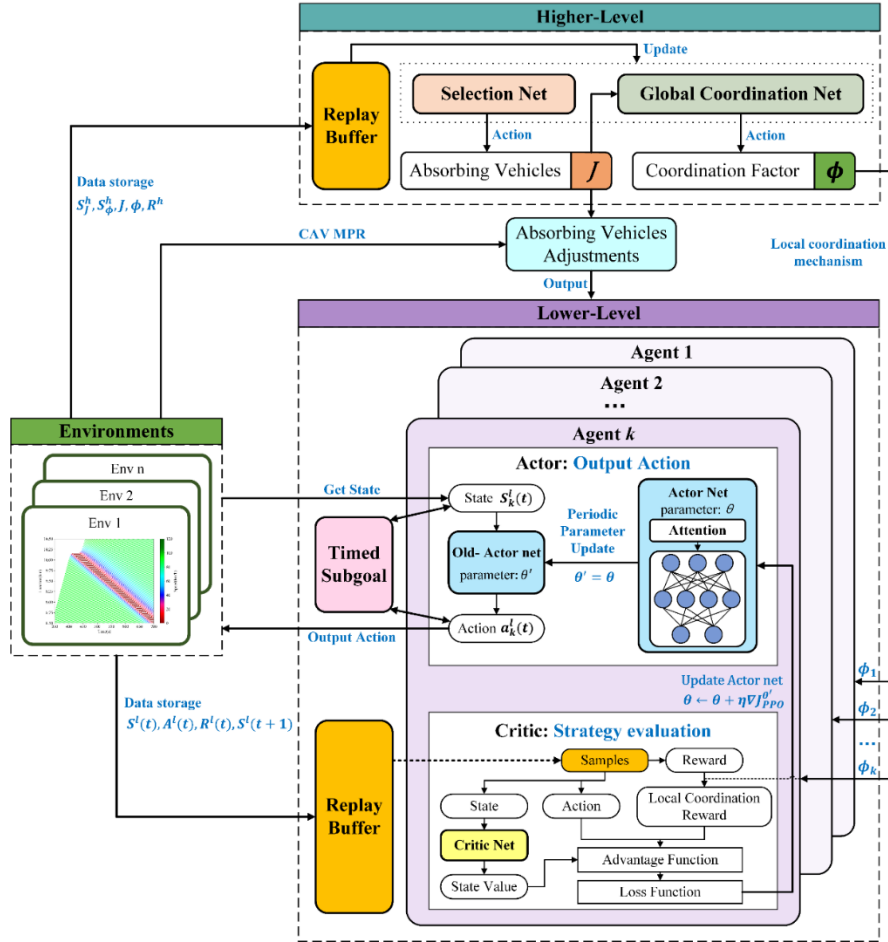


Fig. 10. Multi-Granularity Hierarchical Multi-Agent Coordination Policy Optimization.

The higher-level consists of a selection net and a coordination net, which operate at a coarser temporal resolution. At the beginning of the control process, the higher-level agent determines the absorbing vehicles and formulates the global coordination strategy. These decisions are then distributed to each lower-level agent, guiding them in generating specific control policies and coordination plans. In contrast, the lower-level agents operate at a finer temporal resolution. Each agent generates a timed sub-goal based on its corresponding absorbing vehicle and applies acceleration control to that vehicle according to its local observations and the coordination commands received from the higher level.

4.1.2 Higher level

● Selection net

The higher-level selection net determines which absorbing vehicles are to be controlled. Selection net considers the operating states of all vehicles on the road and provides aggregated information to guide the lower-level agents. The higher-level state S_f^h is defined as:

$$S_f^h = [t_p', v(0, t^S), v(1, t^S), \dots, v(N, t^S), s(0, t^S), s(1, t^S), \dots, s(N, t^S)] \quad ()$$

where t_p' denotes the predicted stop time of the first vehicle. By integrating both temporal and spatial information, S_f^h captures the macroscopic traffic conditions at a given supervisory time t^S .

Higher-level action A_f^h focuses on structural optimization of the platoon system by determining the absorbing vehicle set J . Formally, it can be expressed as:

$$A_f^h = J \quad ()$$

● Platoon segmentation

After the absorbing vehicles are selected, the vehicles on the road are naturally divided into $\chi + 2$ platoons. According to the characteristics of their leading vehicles, these platoons can be categorized into three types: the first platoon, the absorbing platoons, and the uncontrolled platoon, as illustrated in Fig. 11.

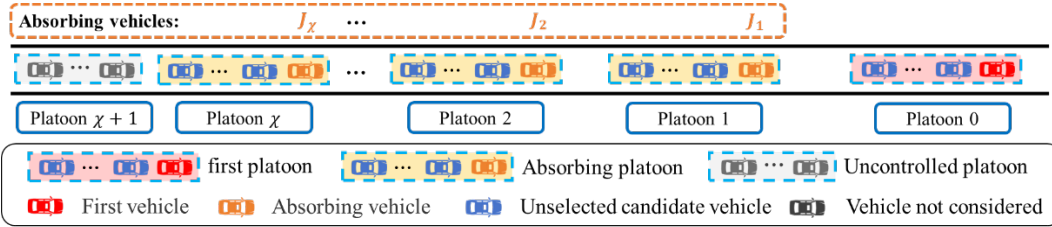


Fig. 11. Platoon segmentation.

The macroscopic features of each platoon are extracted and aggregated to construct its state representation.

State of first platoon $S_0^p(t)$:

$$S_0^p(t) = [v(0, t), \bar{v}_0^c(t), \sigma_0^v(t), v_0^{\min}(t), v_0^{\max}(t), \bar{s}_0^c(t), \sigma_0^s(t), s_0^{\min}(t), s_0^{\max}(t)] \quad ()$$

State of absorbing platoon $S_k^p(t)$, $k = 1, 2, \dots, \chi$:

$$S_k^p(t) = \begin{bmatrix} J_k, s(J_k, t), v(J_k, t), E_k^a(t), \bar{v}_k^p(t), \sigma_k^v(t), v_k^{\min}(t), v_k^{\max}(t), \bar{s}_k^p(t), \sigma_k^s(t), \\ s_k^{\min}(t), s_k^{\max}(t) \end{bmatrix} \quad ()$$

State of the uncontrolled platoon $S_{\chi+1}^p(t)$:

$$S_{\chi+1}^p(t) = [\bar{v}_{\chi+1}^p(t), \sigma_{\chi+1}^v(t), v_{\chi+1}^{\min}(t), v_{\chi+1}^{\max}(t), \bar{s}_{\chi+1}^p(t), \sigma_{\chi+1}^s(t), s_{\chi+1}^{\min}(t), s_{\chi+1}^{\max}(t)] \quad ()$$

Where $\bar{v}_k^p(t)$ is average vehicle speed in platoon k at time t ; $\sigma_k^v(t)$ is standard deviation of vehicle speeds within platoon k at time t ; $v_k^{\min}(t)$ and $v_k^{\max}(t)$ denote minimum and maximum vehicle speeds in platoon k at time t , respectively; $\bar{s}_k^p(t)$ is average inter-vehicle gap in platoon k at time t ; $\sigma_k^s(t)$ is standard deviation of inter-vehicle gaps within platoon k at time t ; $s_k^{\min}(t)$ and $s_k^{\max}(t)$ denote minimum and maximum inter-vehicle gaps in platoon k at time t , respectively; and $E_k^a(t)$ denotes the extent of effect on platoon k caused by absorbing vehicle J_k maintaining its current speed at time t .

We use the extent of affect $E_k^a(t)$, to quantify how the current decision of the absorbing vehicle influences vehicles in platoon k . A larger value indicates that the absorbing vehicle's decision has not yet been widely propagated within the platoon, whereas a smaller value suggests that most vehicles have already aligned with the absorbing strategy. Formally,

$$E_k^a(t) = \frac{1}{n_k^p} \sum_{i=J_k}^{J_k+n_k^p} \frac{v(i, t)}{v^0} \left| \frac{v(i, t)^2}{v(J_k, t)^2} - 1 \right| \{ReLU[v(J_k, t-1) - v(J_k, t)] + 1\} \quad ()$$

Where n_k^p is number of vehicles in platoon k . E_k^a is determined by three main components: (i) Higher platoon

speed increases $E_k^a(t)$ via the factor $v(i, t)/v^0$. (ii) Larger speed discrepancies between a follower and the absorbing vehicle raise $E_k^a(t)$ through $\left| \frac{v(i, t)^2}{v(j_k, t)^2} - 1 \right|$. (iii) Deceleration waves generated by the absorbing vehicle amplify the effect through $\text{ReLU}[v(j_k, t-1) - v(j_k, t)]$, which is positive only when the leader slows down. Fig. 12 illustrates that the influence intensifies with both the platoon's speed and the velocity difference between the platoon and the absorbing vehicle; moreover, when the absorbing vehicle decelerates, the impact increases further.

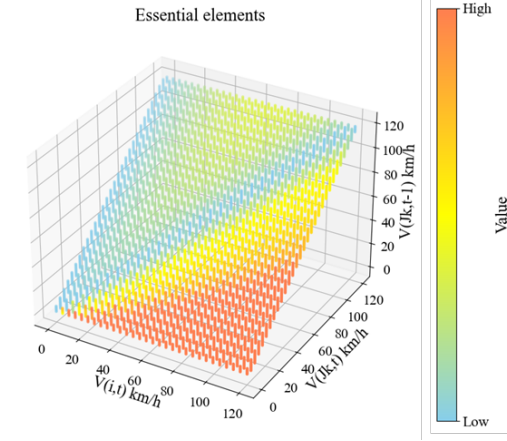


Fig. 12. The extent of affect in different absorbing vehicle decisions.

The platoon reward $r_k^p(t)$ is employed to evaluate the operational state of platoon at each time step. A higher platoon reward indicates a better operational performance. The specific formulation is as follows:

$$r_k^p(t) = -\sum_{i=J_k}^{J_k+n_k^p} Z(i, t) \quad ()$$

● Global coordination net

To prevent absorbing vehicles from focusing excessively on their own platoons, a coordination mechanism is introduced into the proposed algorithm. The higher-level agent employs a global coordination net to assign coordination factors to each lower-level agent. The inputs to the coordination net are as follows:

$$S_\phi^h = [S_0^p(t^s), S_1^p(t^s), \dots, S_\chi^p(t^s), S_{\chi+1}^p(t^s)] \quad ()$$

The outputs of net are as follows:

$$A_\phi^h = \phi = [\phi_1, \phi_2, \dots, \phi_\chi] \quad ()$$

Here ϕ denotes the set of coordination factors, and ϕ_k is the coordination factor assigned to agent k . In this algorithm, coordination factor ϕ_k represents the degree of concern platoon k pays to platoons behind it.

● Higher-level reward

To achieve the optimization objective, the rewards are defined as the negative of the objective function. The higher level focuses on optimizing the entire road scenario, and the higher-level reward R^h can be expressed as the sum of the overall optimization objectives of all vehicles.

$$R^h = \sum_{t=t^s}^T \sum_{k=0}^{\chi+1} r_k^p(t) \quad ()$$

4.1.3 Lower level

During each control step, each lower-level agent selects an action according to its input state. The lower-level local state includes the current platoon's state as well as the states of all platoons ahead.

$$S^l(t) = [S_1^l(t), S_2^l(t), \dots, S_\chi^l(t)] \quad ()$$

$$S_k^l(t) = [S_0^p(t), S_1^p(t), \dots, S_k^p(t)] \quad ()$$

Where $S^l(t)$ is state set of all lower-level agents at time t , and $S_k^l(t)$ is state of the lower-level agent k at time

t .

The action output by each lower-level agent at each control step corresponds to the acceleration command of the absorbing vehicle at that moment.

$$A^l(t) = [a_1^l(t), a_2^l(t), \dots, a_k^l(t), \dots] \quad ()$$

$$a_k^l(t) = a_j(J_k, t), t \in T \quad ()$$

Where $A^l(t)$ is action set of all lower-level agents at time t , and $a_k^l(t)$ action output by lower-level Agent k at time t .

The lower-level agents adopt a local coordination reward, which incorporates the objective functions of the current platoon and other platoons through coordination factors.

$$R^l(t) = [r_1^l(t), r_2^l(t), \dots, r_k^l(t), \dots] \quad ()$$

Where $R^l(t)$ is reward set of all lower-level agents at time t , and $r_k^l(t)$ is local coordination reward of platoon k at time t .

● Local coordination

This study introduces a local coordination mechanism to regulate the rewards of lower-level agents, aiming to balance the global objective with the individual platoon-level rewards of the absorbing vehicles (Fig. 13).

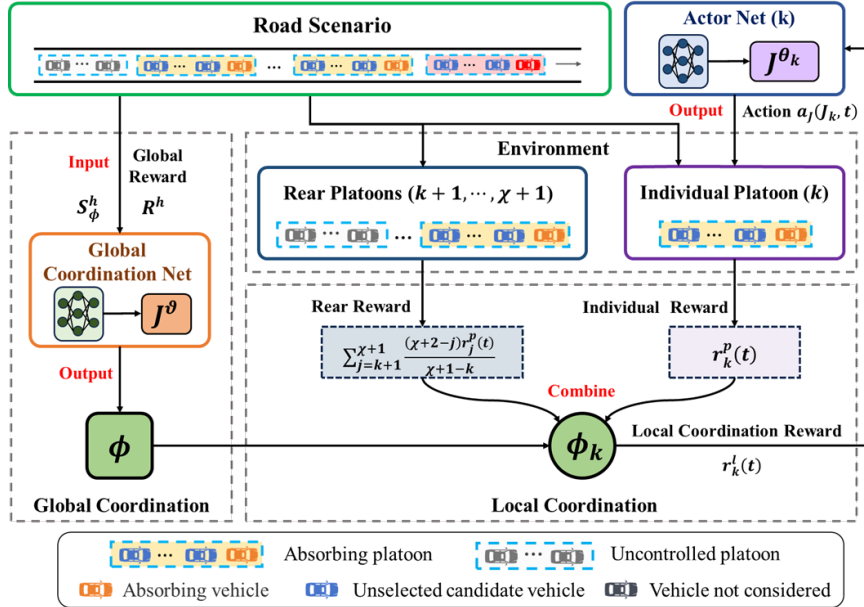


Fig. 13. Coordination mechanism.

By using the coordination factor ϕ_k to adjust reward of lower-level agents, the target platoon k is coordinated with its rear platoons. After local coordination is applied through the coordination factors, each absorbing platoon receives its corresponding local coordination reward.

$$r_k^l(t) = (1 - \phi_k)r_k^p(t) + \phi_k \sum_{j=k+1}^{\chi+1} \frac{(\chi+2-j)r_j^p(t)}{\chi+1-k} \quad ()$$

● Timed subgoal

To accommodate the varying control durations required by different vehicles, this study implements a timed-subgoal mechanism (Fig. 14). Each agent operates within a specific control horizon, after which it transitions into "Observer mode". In this phase, the agent continues to process state inputs and generate virtual actions ($a_{virtual}$), which interact with the environment in a virtual manner without being physically executed. This mechanism ensures continuous data synchronization across the multi-agent system. The resulting interaction data is stored in the replay

buffer with the format $D^l = \{S^l(t), A^l(t), R^l(t), S^l(t+1)\}$, capturing the states, actions, rewards, and subsequent states for training.

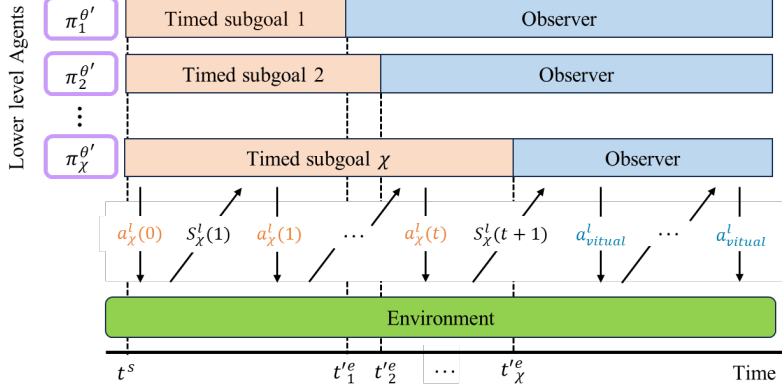


Fig. 14. Timed subgoal.

4.2 Training objectives

4.2.1 Network Architecture

The network serves as the decision-making module, mapping local observations to vehicle acceleration commands. To balance computational efficiency and generalization capability, this study adopts two distinct implementations: a lightweight self-attention network for standard scenarios and a pre-trained time-series foundation model for complex environments.

In scenarios prioritizing efficiency, the network incorporates a self-attention layer to extract correlations among variables from the input state (Vaswani et al., 2017). The computation is formalized as follows:

$$\begin{aligned} Q &= W^Q I; K = W^K I; V = W^V I & () \\ O &= V \cdot \text{softmax}(K^T Q) & () \end{aligned}$$

Where I denotes the input vector of the attention layer. W^Q , W^K , and W^V are the weight matrices for the Query (Q), Key (K), and Value (V) vectors, respectively. The output O is generated by weighting V based on the attention scores derived from the dot product of Q and K , enabling the agent to focus on critical interactions within the traffic stream.

For complex scenarios requiring robust temporal modeling, the network is fine-tuned using open-source pre-trained time-series foundation models, specifically TimesFM (Das et al., 2024) and Timer (Liu et al., 2024). TimesFM utilizes a Transformer-based architecture to enable zero-shot forecasting, while Timer employs a physics-inspired continuous-dynamics formulation for enhanced stability. Although incorporating these foundation models increases training costs, their superior ability to capture long-term dependencies significantly enhances the agent's generalization performance in dynamic traffic environments.

4.2.2 Loss function

The higher-level agent is responsible for the structural optimization of the platoon system, specifically selecting the absorbing vehicles (J) and determining the coordination factor (ϕ). The objective is to maximize the expected global reward R^h while encouraging exploration through policy entropy. The total loss function L^θ for the higher-level agent is defined as:

$$L^\vartheta = \lambda_1 R^h \pi^\vartheta(J, \phi | S_f^h, S_\phi^h) - \lambda_2 \sum_a \pi^\vartheta(a | S_f^h, S_\phi^h) \log \pi^\vartheta(a | S_f^h, S_\phi^h) \quad ()$$

Where π^ϑ represents the policy of the higher-level network parameterized by ϑ . The first term maximizes the global reward, while the second term represents the entropy of the policy, weighted by λ_2 , to prevent premature convergence and encourage diverse strategies.

The lower-level agents optimize the specific acceleration controls for the absorbing vehicles. The training process utilizes an actor-critic architecture involving an actor net (π_k^θ), an old actor net ($\pi_k^{\theta'}$), and a critic net.

The critic net estimates the state value $V_k^\varphi[S_k^l(t)]$ to evaluate the current policy. The parameters φ are updated to minimize the difference between the predicted value and the actual target value using the temporal difference error:

$$V_k^\varphi[S_k^l(t)] \leftarrow V_k^\varphi[S_k^l(t)] + \alpha \{r_k^l(t) + \gamma V_k^\varphi[S_k^l(t)] - V_k^\varphi[S_k^l(t)]\} \quad ()$$

Here, α is the learning rate of the Critic Net, and γ is the discount factor.

The overall objective function for the lower-level actor net, L^{θ_k} , composite of three distinct loss components and a Kullback-Leibler (KL) divergence constraint:

$$L^{\theta_k} = \lambda_1 L_1^{\theta_k} + \lambda_2 L_2^{\theta_k} + \lambda_3 L_3^{\theta_k} - KL(\pi_k^\theta, \pi_k^{\theta'}) \quad ()$$

The components are defined as follows:

- **Minimized Approximation Loss ($L_1^{\theta_k}$)**

This term employs a clipped advantage function to ensure stable policy updates. It limits the change in the policy ratio $\varepsilon(t) = \frac{\pi_k^\theta[a_k^l(t) | S_k^l(t)]}{\pi_k^{\theta'}[a_k^l(t) | S_k^l(t)]}$ to a trusted region defined by ϵ :

$$L_1^{\theta_k} = \sum_t \min\{\varepsilon(t) A_k^{\theta'}[S_k^l(t), a_k^l(t)], \text{clip}[\varepsilon(t), 1 - \epsilon, 1 + \epsilon] A_k^{\theta'}[S_k^l(t), a_k^l(t)]\} \quad ()$$

Where $A_k^{\theta'}$ is the advantage function evaluate the quality of action $a_k^l(t)$.

- **Policy Entropy Loss ($L_2^{\theta_k}$)**

This term encourages exploration by preventing the policy from becoming overly deterministic:

$$L_2^{\theta_k} = - \sum_t \int \pi_k^\theta[a | S_k^l(t)] \log \pi_k^\theta[a | S_k^l(t)] da \quad ()$$

- **Knowledge Distillation Loss ($L_3^{\theta_k}$)**

To avoid local optima and enhance performance, the agent distills knowledge from a teacher policy π_k^S , defined as the optimal policy previously generated by the agent:

$$L_3^{\theta_k} = - \sum_t \int \pi_k^S[a | S_k^l(t)] \log \pi_k^\theta[a | S_k^l(t)] da \quad ()$$

- **KL Divergence**

To prevent excessive deviation between the current policy π_k^θ and the old policy $\pi_k^{\theta'}$, the KL divergence is subtracted from the objective:

$$KL(\pi_k^\theta, \pi_k^{\theta'}) = \beta \sum_t \int \pi_k^\theta[a | S_k^l(t)] \log \frac{\pi_k^\theta[a | S_k^l(t)]}{\pi_k^{\theta'}[a | S_k^l(t)]} da \quad ()$$

The weight coefficient β is dynamically adjusted according to whether the KL divergence falls outside the target range (KL_{min}, KL_{max}).

4.3. Hyperparameters and Convergence Effects

The hyperparameters adopted in this study are summarized in [Table 1](#). All algorithms considered in this

study are implemented in Python to ensure the reproducibility of parallel sampling, replay-buffer management, and network-update processes. For the hyperparameter configuration of the core algorithms, we followed the benchmark recommendations of Schulman et al. (2017) to ensure algorithmic stability and convergence in continuous-control tasks.

Table 1. Hyperparameter values.

Parameters	Values	Parameters	Values
Number of parallel environments (M)	4	Action net learning rate (η)	0.0003
Maximum training batch (E)	150	Strategy ratio clipping factor (ϵ)	0.2
Hidden layer number in the Actor Net	4	Critic net learning rate (α)	0.0003
Hidden layer number in the Critic Net	2	Discount factor (γ)	0.99
Number of neurons in hidden layers	64	Global reward weight (λ_1)	1
Batch size of Higher-level replay buffer	32	Policy entropy weight (λ_2)	0.02
Batch size of Lower-level replay buffer	624	Knowledge distillation weight (λ_3)	0.005
Discount rate (κ)	2	KL divergence weight (β)	0.5

The reward–episode curves show that the proposed algorithm converges well under all settings with one to four absorbing vehicles (Fig. 15). In all cases, the cumulative reward rises rapidly at the beginning and then gradually levels off with reduced fluctuations, indicating rapid learning in the early stage and stable policy convergence thereafter. As the number of absorbing vehicles increases, convergence becomes slightly slower, and the final reward varies modestly, reflecting the greater coordination difficulty in higher-dimensional action spaces. Nevertheless, even with four absorbing vehicles, the algorithm reaches a stable plateau within approximately 110–150 episodes, demonstrating satisfactory convergence speed and robustness.

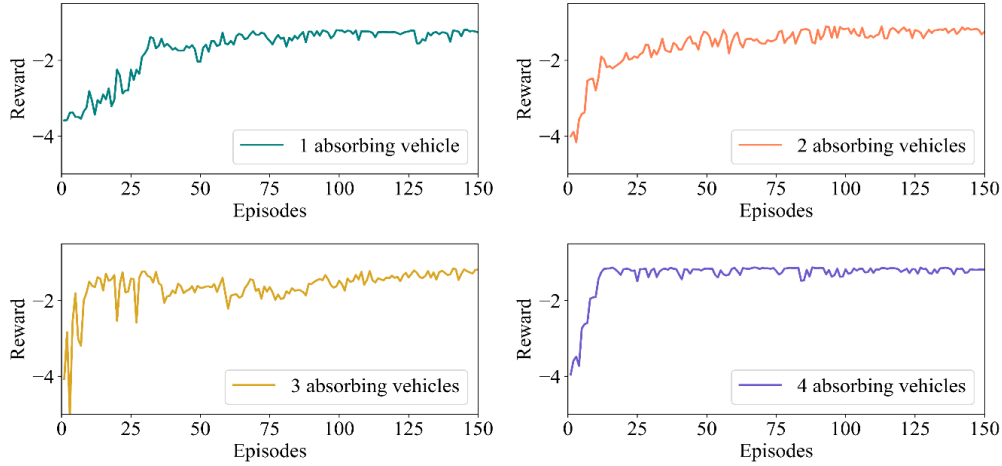


Fig. 15. Convergence curve.

Moreover, the curves under different maximum numbers of absorbing vehicles show similar fluctuation amplitudes in the stable stage, without clear differences in convergence speed. This suggests that scaling the control size from one to four vehicles does not compromise training stability or optimization consistency. Overall, the model achieves stable learning and effective convergence across different control scales, indicating good robustness and transferability.

5. Numerical analysis

5.1 Scenario information

Two test scenarios are constructed to evaluate the control policy. The first serves for training with 100% CAV penetration and is designed to verify whether the policy can reproduce its performance under identical, deterministic conditions. The second investigates robustness by varying the CAV penetration rate (e.g., 5%–50%) under random spatial distributions to examine how control effectiveness scales in mixed traffic.

All simulations are conducted on a 12 km single-lane freeway segment with 300 vehicles to represent near-capacity traffic oscillations; uniform road conditions are assumed, and external disturbances are ignored. Vehicles initially travel at a constant speed of $v_0 = 90$ km/h, with initial space headways uniformly distributed between 30 m and 40 m; the traffic stream consists of mixed CAV and HDV traffic, in which CAVs receive real-time control commands via roadside units and HDVs follow IDM. IDM parameters are adopted from prior literature with clear physical interpretations: maximum comfortable acceleration $a_{max} = 1.4$ m/s², maximum comfortable deceleration $b_{max} = 2$ m/s², HDV desired time headway $T_{d,HDV} \in [0.6, 1.6]$ s, CAV desired time headway $T_{d,CAV} = 1.1$ s, minimum standstill spacing $s_0 = 2$ m, vehicle length $L = 5$ m, free-flow acceleration exponent $\delta = 4$, and desired speed v_d set by road type (e.g., 80–100 km/h for urban expressways and 120 km/h for highways).

The traffic oscillation is triggered by the lead vehicle acting as a "jam maker" at the 10 km mark through a four-stage process (Fig. 16). After a period of stable flow (Stage 0), the leader decelerates to a complete stop (Stage 1) and remains stationary for a brief period (Stage 2), causing local density to increase sharply. Finally, the leader accelerates back to the initial speed (Stage 3), thereby propagating a stop-and-go shockwave upstream through the platoon, which serves as the primary disturbance for testing the control policy.

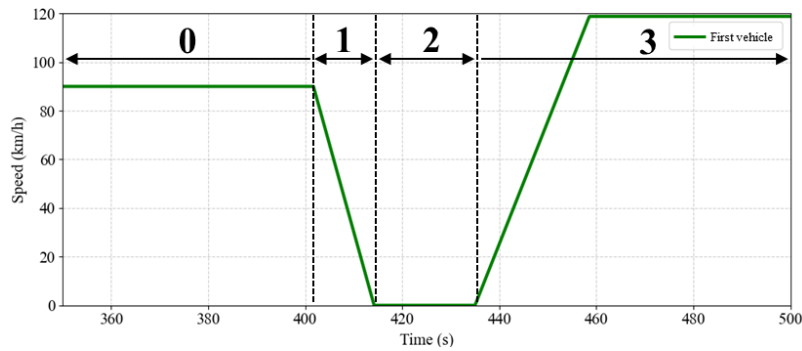


Fig. 16. Speed profile of first vehicle.

The selection of these parameters ensures smooth deceleration and acceleration processes, allowing the generated traffic oscillations to propagate stably in the upstream direction without inducing model instability due to excessive deceleration or overly rapid acceleration (Fig. 17).

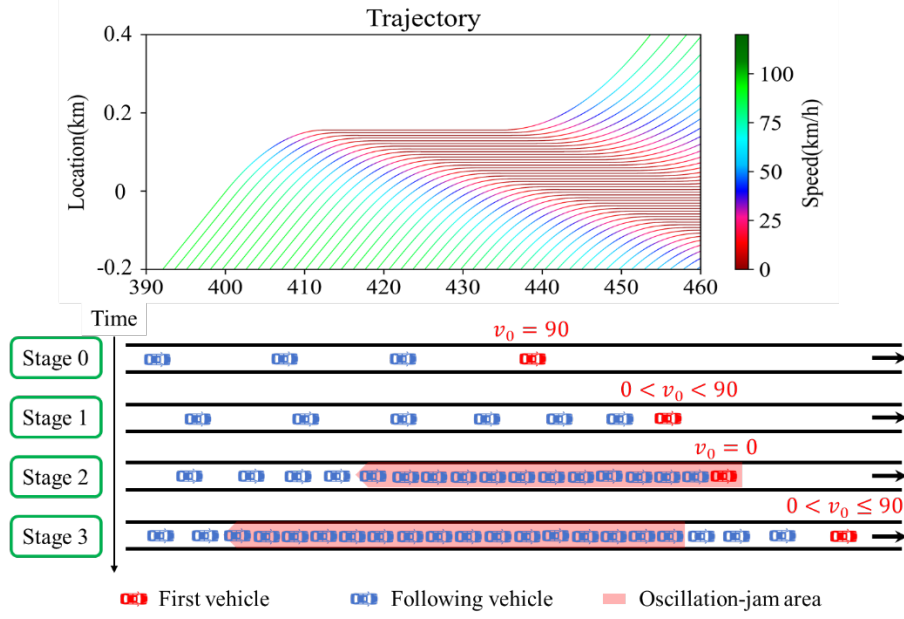
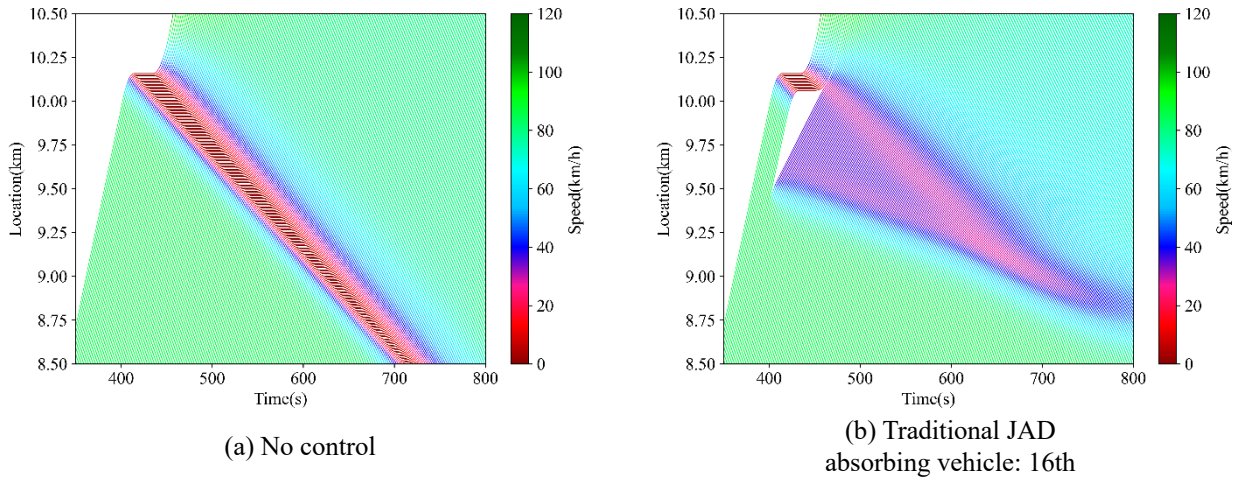


Fig. 17. Schematic diagram of oscillation formation.

5.2 Result analysis

Fig. 18 presents the microscopic space–time trajectories under the uncontrolled scenario, the conventional single-vehicle JAD strategy (Zheng et al., 2020), and the proposed cooperative JAD strategies with different numbers of controlled vehicles. In the uncontrolled case, the disturbance generates a pronounced low-speed congestion region that propagates upstream and significantly disrupts the following traffic. The conventional single-vehicle JAD strategy can only partially reduce congestion because the control capability of a single vehicle is inherently limited. By contrast, the cooperative JAD strategy suppresses traffic oscillations more effectively, and its performance improves as the number of controlled vehicles increases. In particular, the four-vehicle cooperative case almost eliminates the congested region and quickly restores traffic flow to a smooth free-flow state, demonstrating the clear advantage of cooperative control in damping shockwave propagation and enhancing traffic stability.



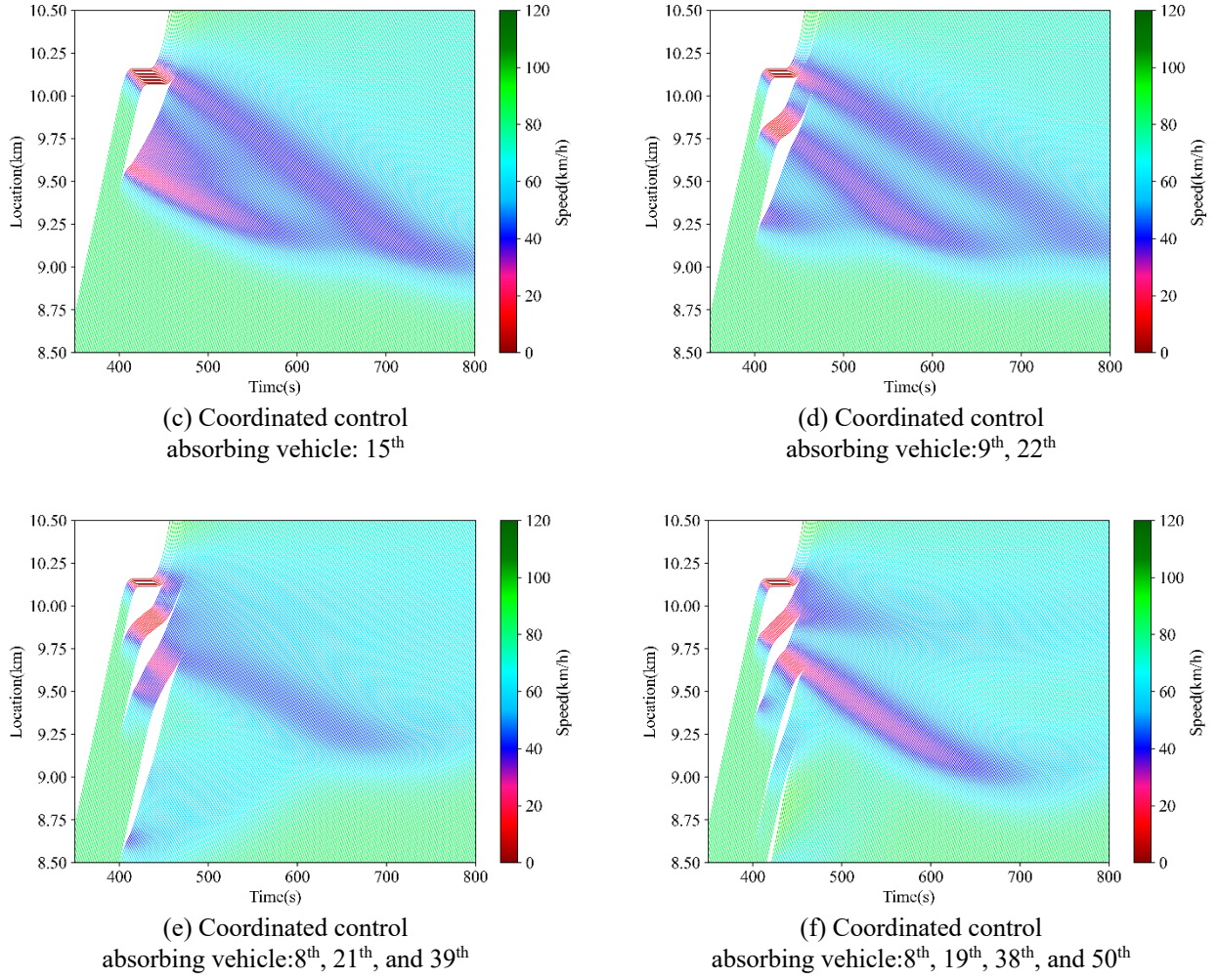


Fig. 18. Simulated trajectory diagram.

5.2.1 Performance measurements

To quantify the effectiveness of JAD control from multiple perspectives, this study adopts four categories of performance measures: surrogate safety, efficiency, driving comfort, and fuel consumption, all of which are computed by aggregating vehicle-level quantities over the simulation horizon. For surrogate safety, the TIT and TET are defined as follows:

$$TIT = \sum_{t=0}^T \sum_{i=1}^N TIT(i, t) \Delta t \quad ()$$

where i and t denote the vehicle index and time index, respectively, N is the total number of vehicles, T is the simulation horizon, and Δt is the time step.

$$TET = \sum_{t=0}^T \sum_{i=1}^N TET(i, t) \Delta t \quad ()$$

$$TET(i, t) = \begin{cases} 1, & \forall TTC(i, t) < TTC_0 \\ 0, & else \end{cases} \quad ()$$

where $TET(i, t)$ indicates whether the TTC of vehicle i at time t is less than the TTC threshold, $TTC(i, t)$ is the time-to-collision of vehicle i at time t , and TTC_0 is the TTC threshold.

In addition to safety, traffic efficiency is evaluated using total delay and total stop time. The total delay is computed as:

$$D = \sum_{t=0}^T \sum_{i=1}^N d(i, t) \quad ()$$

where D denotes the total delay and $d(i, t)$ denotes the delay of vehicle i at time t .

The total stop time is measured by:

$$T_s = \sum_{t=0}^T \sum_{i=1}^N t_s(i, t) \Delta t \quad ()$$

$$t_s(i, t) = \begin{cases} 1, & \text{if } v(i, t) < 0.5 \text{ m/s} \\ 0, & \text{if } v(i, t) \geq 0.5 \text{ m/s} \end{cases} \quad ()$$

where T_s denotes the total stop time and $t_s(i, t)$ indicates whether vehicle i has stopped at time t , and $v(i, t)$ is the speed of vehicle i at time t .

Driving comfort is approximated using the standard deviation of vehicle speeds:

$$C = \sqrt{\frac{1}{T} \sum_{t=0}^T \sum_{i=1}^N (v(i, t) - \bar{v})^2} \quad ()$$

where C is the comfort metric replaced by the standard deviation of vehicle speeds, and $\bar{v}$ is the average of all instantaneous speeds of all vehicles from entering to exiting the test road section.

Fuel consumption is evaluated using the following aggregate fuel-consumption model:

$$F = \sum_{t=0}^T \sum_{i=1}^N a_0 + a_1 v(i, t) + a_2 v(i, t)^2 + a_3 a(i, t) v(i, t) \quad ()$$

where F denotes total fuel consumption, $a(i, t)$ and $v(i, t)$ are the acceleration and speed of vehicle i at time t , and the coefficients are given by $a_0 = 0.1569$, $a_1 = 0.0020$, $a_2 = 1.13 \times 10^{-4}$, and $a_3 = 0.0003$.

5.2.2 Evaluation results

To comprehensively evaluate the proposed strategy at both the macroscopic and microscopic levels, this section analyzes total delay, total stop time, collision-risk exposure, comfort, fuel consumption, and the weighted overall objective based on the results summarized in Table 2. The uncontrolled scenario exhibits severe congestion, high delay, and elevated safety risk, whereas the conventional single-vehicle JAD strategy reported by Zheng et al. (2020) improves traffic performance only to a limited extent. By contrast, the proposed cooperative JAD strategy achieves consistent improvements in efficiency, safety, comfort, and fuel economy, and its performance improves progressively as the number of controlled vehicles increases. In particular, the four-vehicle cooperative case yields the lowest overall objective value, indicating that distributed cooperative control is more effective than single-vehicle control in suppressing traffic oscillations and balancing multiple optimization goals.

Table 2. Evaluation results.

Control type	No control	Traditional JAD	Coordinated control			
χ	0	1	1	2	3	4
TET	370	36	34	27	23	20
(%)	-	90.27	90.81	92.7	93.78	94.59
TIT	6.26	2.85	2.51	2.33	2.16	2.06
(%)	-	54.47	59.9	62.78	65.5	66.29
D	10702	9286	9056	8761	8456	8276
(%)	-	13.23	15.38	19.07	21.92	22.67
T_s	4769	322	301	174	153	153
(%)	-	93.25	93.69	96.35	96.79	96.79
C	6615	6062	5847	5614	5453	5333

	(%)	-	39	44	51	55	57
F	(L)	9.62	9.2	9.02	8.64	8.18	8.15
	(%)	-	4.36	6.24	10.18	14.97	15.28
Z	-	2	1.32	1.25	1.18	1.13	1.11
	(%)	-	33.85	37.64	40.93	43.71	44.48

To further examine the generalization capability of the proposed framework in more complex scenarios, the traffic-oscillation trigger time was extended from a fixed setting to a random interval of 20–60 s, thereby introducing greater uncertainty into both disturbance onset and evolution. Under this generalized setting, two representative open-source pre-trained time-series models, TimesFM and Timer, were adopted as backbone action networks and fine-tuned using LoRA+RL. As shown in Table 3, all models achieved significant improvements over the uncontrolled baseline across different numbers of controlled vehicles. Moreover, the performance differences among the models remain consistently small, indicating that the proposed MG-HiMaPO framework is highly compatible with different backbone architectures and relatively insensitive to model size. These results suggest that pre-trained time-series models, after lightweight adaptation, can effectively support cooperative JAD control and provide robust generalization in traffic scenarios with randomized oscillation disturbances.

Table 3. Average optimization effectiveness across different models.

Base model	Parameter scale	χ	$\bar{D}$		$\overline{TET}$		$\overline{T_s}$	
			(s)	(%)	-	(%)	(s)	(%)
baseline	-	No control	12731	-	509	-	5442	-
TimesFM-1.0	0.2B	3	9202	27.72	45	91.16	338	93.79
		4	9043	28.97	22	95.68	281	94.84
TimesFM-2.5	0.2B	3	9231	27.49	39	92.34	338	93.79
		4	9018	29.19	23	95.48	279	94.87
Timer	0.08B	3	9173	27.95	40	92.14	338	93.79
		4	9026	29.10	23	95.48	280	94.85
Timer	0.13B	3	9192	27.80	40	92.14	338	93.79
		4	9052	28.90	24	95.28	276	94.93

5.3 Sensitivity analysis

5.3.1 Sensitivity Analysis of State Variables

To examine the influence of each state variable on the decisions of the reinforcement-learning agents, SHAP (Lundberg and Lee, 2017) is adopted for sensitivity analysis. For a given feature i , its SHAP value is defined as

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N|-|S|-1)!}{|N|!} [f(S \cup \{i\}) - f(S)] \quad ()$$

where ϕ_i is the contribution of feature i , N is the set of all input features, S is any subset of $N \setminus \{i\}$, and $f(S \cup \{i\}) - f(S)$ denotes the marginal contribution of feature i . Using the four-absorbing-vehicle case as an example, the SHAP values of all state variables are computed to characterize their overall influence distributions.

As shown in Fig. 19, spacing to the lead vehicle, average speed, and minimum speed are the most influential

variables for all agents, with minimum speed exhibiting the widest SHAP-value range, especially for Agents 3 and 4. This suggests that the lower bound of local traffic speed strongly shapes acceleration and deceleration decisions. In contrast, lead-vehicle speed and maximum speed have smaller effects, while absorbing-vehicle ID, speed variance, and spacing variance remain near zero, indicating that the learned policy relies primarily on physical traffic states rather than vehicle identity or minor fluctuations. Although the overall patterns are similar across agents, some differences remain: Agent 1 is more consistently affected by lead-vehicle spacing, whereas Agent 4 is more sensitive to minimum speed.

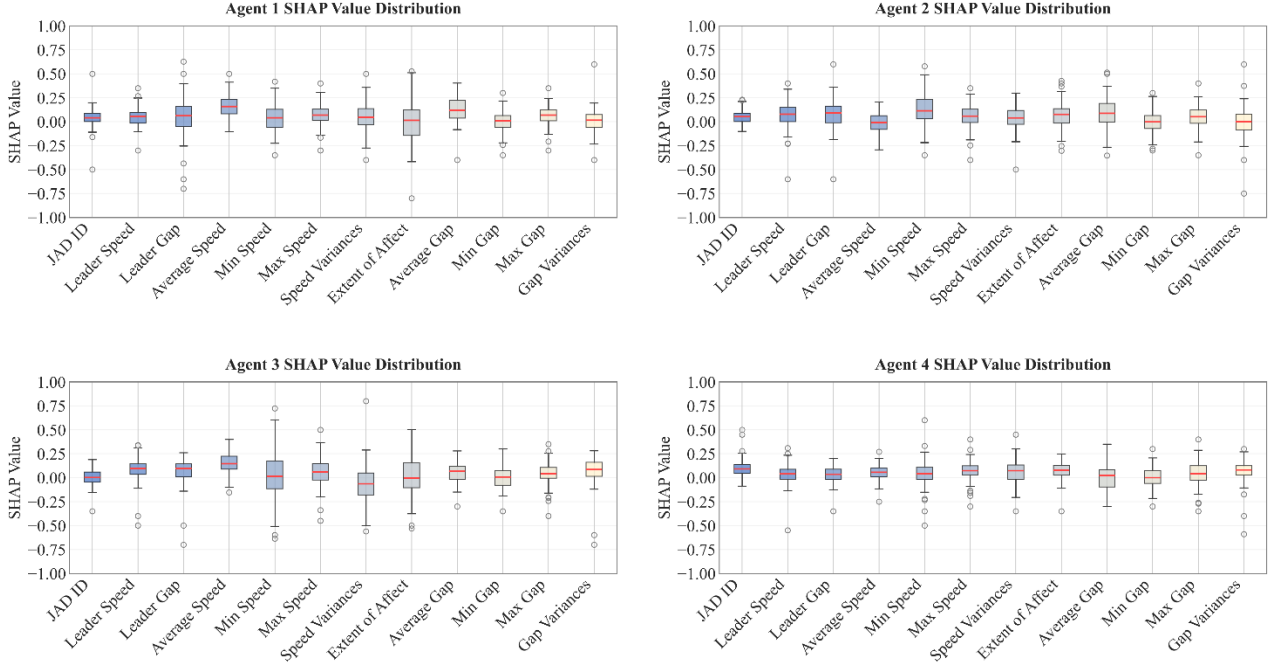


Fig. 19. Variable importance.

5.3.2 Impact of Penetration Changes on Strategy

To evaluate the adaptability and robustness of the cooperative JAD strategy under different CAV market penetration rates (MPRs), this section conducts a set of incremental experiments in which the CAV penetration rate increases from 5% to 50%, a setting commonly used in mixed-traffic studies to assess the influence of CAV share on traffic performance. As summarized in Table 4, all major performance indicators improve progressively with increasing penetration rate, indicating a clear positive relationship between control effectiveness and the proportion of controllable vehicles, which is consistent with prior findings in mixed CAV traffic research.

Table 4 Performance metrics statistics at different CAV MPR.

CAV MPR	D (s)	T_s (s)	C	TET	TIT	Z
5%	8954	251	1984621	31	2.74	1.27
10%	8732	191	1818541	27	2.59	1.23
20%	8603	179	1754133	25	2.4	1.19
30%	8499	165	1684133	23	2.24	1.15
40%	8415	161	1635846	23	2.11	1.12

The microscopic space–time trajectories in Fig. 20 further illustrate the effectiveness of the strategy at low penetration rates. Even under the extremely low penetration rate of 5%, the traffic wave is visibly weakened, and the width of the congested region is reduced. When the penetration rate increases to 10%, the suppression effect becomes much more pronounced: the congested region dissipates faster, and the trajectories become smoother and more uniform, indicating a substantial reduction in speed fluctuations. These results show that even a small proportion of CAVs can, through cooperative control, disrupt and attenuate the propagation of congestion waves.

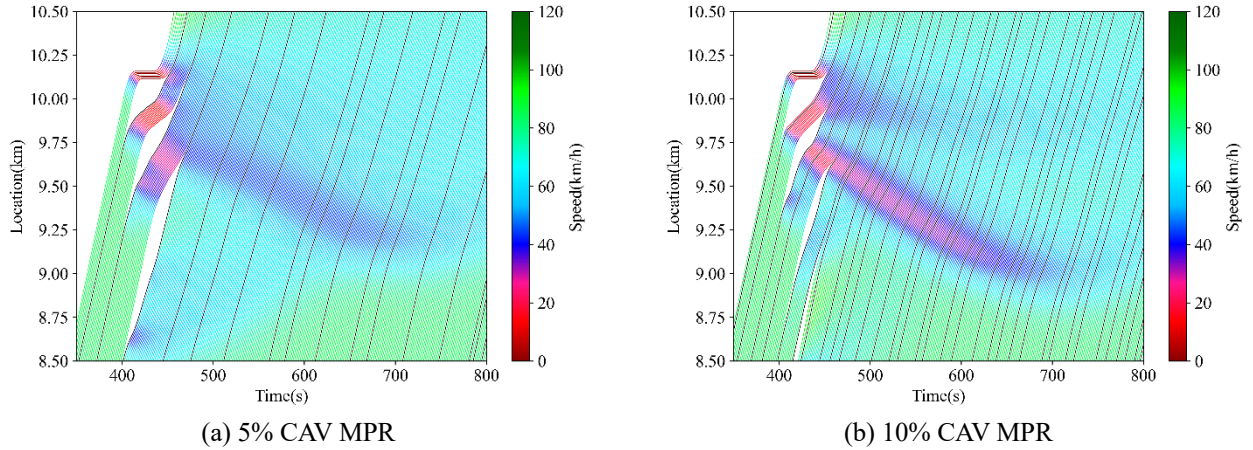
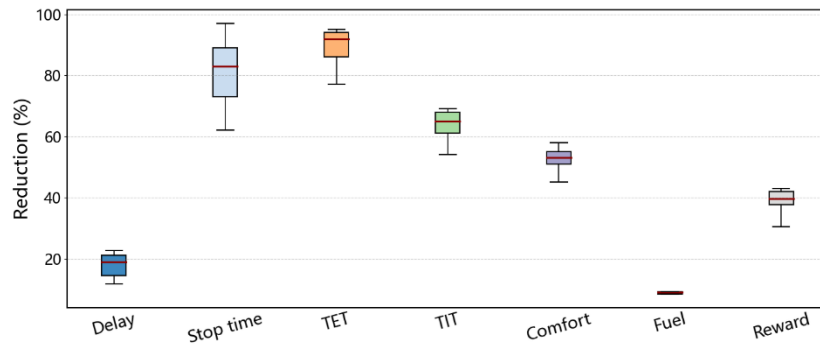


Fig. 20. Simulation results at 5% and 10% CAV MPR.

Fig. 21 provides a more detailed comparison of the improvement rates under the 5% and 10% penetration scenarios. The results show that the cooperative JAD strategy already delivers considerable marginal benefits at the early stage of CAV deployment, especially in reducing stop time and safety risk. When the penetration rate increases from 5% to 10%, the improvements in all indicators become stronger and more stable, with particularly notable gains in stop time and comfort. Although the reductions in delay and fuel consumption are relatively smaller, they still show a consistently positive trend.



(a) 5% CAV MPR

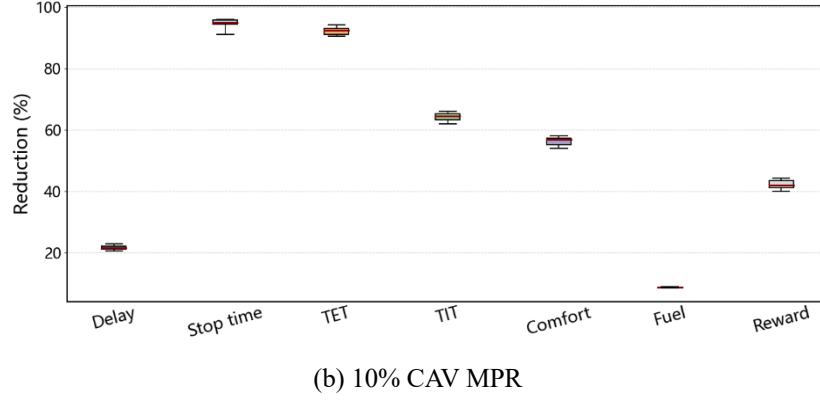


Fig. 21. Comparison of indicator optimization rates at 5% and 10% CAV MPR.

5.4 Extended Research: The Impact of Lane Changing Behavior on Strategy

In the previous modeling and simulation settings, the analysis was restricted to a single-lane scenario in order to focus on the mechanism by which the JAD strategy suppresses traffic oscillations in the longitudinal spatiotemporal domain. To this end, the two-lane freeway scenario considered in this section adopts the widely used LC2013 lane-changing model (Erdmann, 2015). In this scenario, HDVs follow the default LC2013 parameter settings to reproduce realistic lane-changing disturbances, while CAVs implement the proposed cooperative JAD strategy in the two-dimensional mixed-traffic environment. A lead-vehicle deceleration disturbance is introduced in each lane.

The results show that, under the ideal condition of 100% CAV penetration, the cooperative JAD strategy still achieves good control performance in the two-lane scenario, as shown in Table 5. Even in the presence of lane-changing behavior, multi-vehicle longitudinal cooperation can effectively mitigate traffic oscillations when random selfish disturbances from HDVs are absent.

Table 5 Optimization results in multi-lane scenarios.

Control type	χ	TET	TIT	T_s	D
No control	0	856	10.56	10314	22715
	2	148	5.42	1200	18975
Coordinated control	3	132	5.31	1143	18140
	4	131	5.43	1021	18422

A markedly different trend emerges in mixed traffic with lower CAV penetration rates. As the proportion of HDVs increases, frequent and selfish lane changes continuously disrupt the buffer gaps intentionally created by the absorbing vehicles. As a result, the smoothing process executed by the JAD vehicles is repeatedly interrupted, and the original slow-in-fast-out control pattern becomes difficult to maintain.

Both the simulation results and the macroscopic observations indicate that, once the HDV lane-changing frequency exceeds a critical level, the performance of the cooperative JAD strategy deteriorates sharply, as illustrated in Fig. 22. The buffer space originally reserved for damping longitudinal oscillations is consumed by lane-changing interference, and repeated acceleration-deceleration reconstruction may even induce secondary disturbances. These findings suggest that, in realistic mixed traffic, relying solely on one-dimensional longitudinal coordination is insufficient and that future traffic control should move toward an integrated two-dimensional framework that combines longitudinal speed smoothing with lateral lane-change guidance.

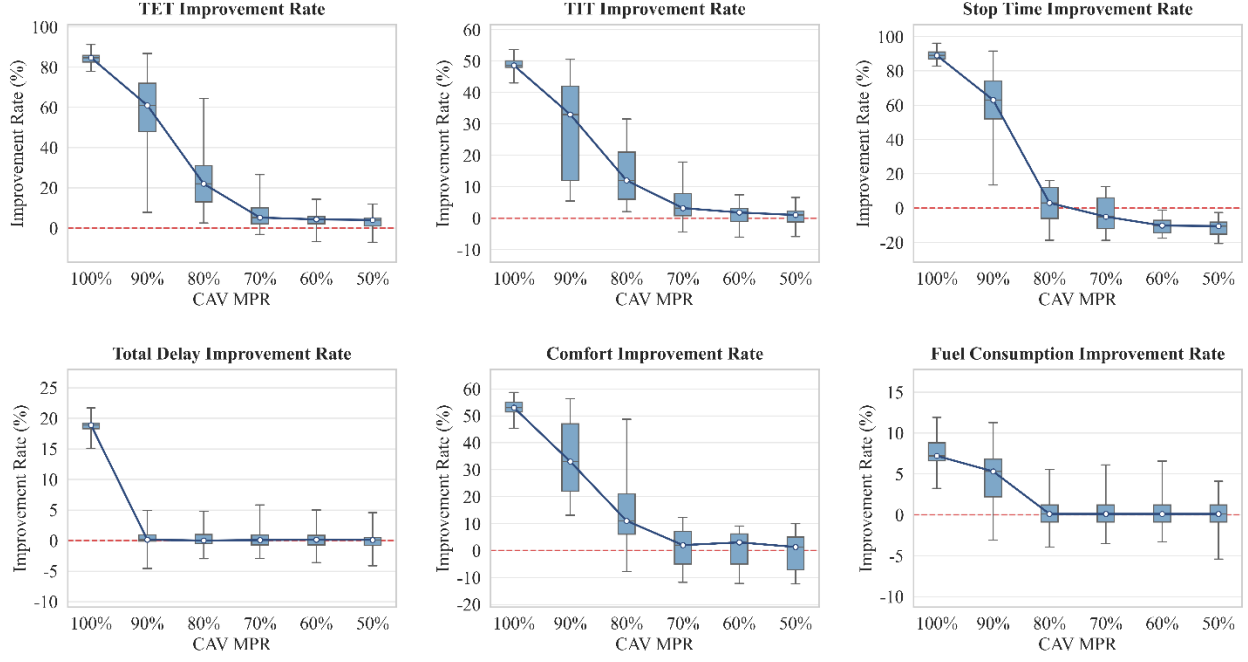


Fig. 22. Optimization results under different CAV MPR in multi-lane scenarios

6. Conclusions

Compared with previous JAD studies, the present study makes two main contributions. First, it moves beyond the predominantly single-vehicle, constant-speed control paradigm adopted in earlier studies by developing a real-time multi-vehicle coordinated JAD framework based on hierarchical reinforcement learning, which integrates vehicle selection, coordination, and trajectory control into a closed-loop decision process. Second, it complements prior simulation-oriented studies with a fundamental-diagram-based interpretation of when and why JAD can improve traffic efficiency, thereby providing a clearer theoretical explanation of the efficiency-recovery mechanism.

The effectiveness of the proposed method is further supported by the comparative results in Table 6: relative to the strategies of He et al. (2017) and Zheng et al. (2020), our optimal strategy achieves the best performance across all reported indicators, including TET (94.6%), total stop time T_s (96.8%), delay D (22.7%), fuel consumption F (15.28%), comfort C (57%), and the overall weighted objective Z (44.5%). These results indicate that the proposed coordinated multi-vehicle control strategy not only outperforms earlier representative methods in safety and efficiency but also delivers more balanced improvements in fuel economy, driving comfort, and overall system performance.

Table 6 Comparison of results

Strategy	TET	T_s	D	F	C	Z
Strategy by He et al. (2017)	85.9 %	81.5 %	8.5 %	8.45 %	35%	29.7 %
Strategy by Zheng et al. (2020)	90.3 %	93.3 %	13.2 %	9.20 %	39%	33.9 %
Our optimal strategy	94.6 %	96.8 %	22.7 %	15.28 %	57%	44.5 %

Reference

- Jerath, K., Gayah, V. V., & Brennan, S. N. (2024). Mitigating delay due to capacity drop near freeway bottlenecks: Zones of influence of connected vehicles. *PloS one*, 19(6), e0301188.
- Knoop, V., & Hoogendoorn, S. (2022). Free flow capacity and queue discharge rate: Long-term changes. *Transportation Research Record*, 2676(7), 483-494.
- Goldmann, K., & Sieg, G. (2021). Quantifying the phantom jam externality: The case of an Autobahn section in Germany. *European transport research review*, 13(1), 12.
- Fageda, X., & Albalade, D. (2020). On the Relationship Between Congestion and Road Safety in Cities. *Available at SSRN 3600139*.
- U.S. Department of Energy. (n.d.). *Driving more efficiently*. Energy Saver. Retrieved January 10, 2026, from <https://www.energy.gov/energysaver/driving-more-efficiently>
- Huang, Y., Ng, E. C., Zhou, J. L., Surawski, N. C., Chan, E. F., & Hong, G. (2018). Eco-driving technology for sustainable road transport: A review. *Renewable and sustainable energy reviews*, 93, 596-609.
- He, H. D., Lu, D. N., Zhao, H. M., & Peng, Z. R. (2024). Characterizing CO₂ and NO_x emission of vehicles crossing toll stations in highway. *Transportation Research Part D: Transport and Environment*, 126, 104024.
- Briani, M., Manzo, R., Piccoli, B., & Rarità, L. (2024). Estimation of NO_x and O₃ reduction by dissipating traffic waves. *Networks & Heterogeneous Media*, 19(2).
- R. Nishi, A. Tomoeda, K. Shimura, and K. Nishinari, "Theory of jamabsorption driving," *Transportation Research Part B: Methodological*, vol. 50, pp. 116–129, 2013.
- R. Nishi, "Theoretical conditions for restricting secondary jams in jamabsorption driving scenarios," *Physica A: Statistical Mechanics and its Applications*, vol. 542, p. 123393, 2020.
- Y. Taniguchi, R. Nishi, T. Ezaki, and K. Nishinari, "Jam-absorptiondriving with a car-following model," *Physica A: Statistical Mechanicsand its Applications*, vol. 433, pp. 304–315, 2015.
- Newell, G. F. (1993). A simplified theory of kinematic waves in highway traffic, part I: General theory. *Transportation Research Part B: Methodological*, 27(4), 281-287.
- Z. He, L. Zheng, L. Song, and N. Zhu, "A jam-absorption driving strategy for mitigating traffic oscillations," *IEEE Transactions on Intelligent Transportation Systems*, vol. 18, no. 4, pp. 802–813, 2017.
- S. Wang, Z. Li, Z. Cao, A. Jolfaei, and Q. Cao, "Jam-absorption driving strategy for improving safety near oscillations in a connected 14 vehicle environment considering consequential jams," *IEEE Intelligent Transportation Systems Magazine*, vol. 14, no. 2, pp. 41–52, 2022.
- Y. Zheng, G. Zhang, Y. Li, and Z. Li, "Optimal jam-absorption driving strategy for mitigating rear-end collision risks with oscillations on freeway straight segments," *Accident Analysis & Prevention*, vol. 135, p. 105367, 2020.
- Y. Han, H. Yu, Z. Li, C. Xu, Y. Ji, and P. Liu, "An optimal control-based vehicle speed guidance strategy to improve traffic safety and efficiency against freeway jam waves," *Accident Analysis & Prevention*, vol. 163, p. 106429, 2021.
- S. Li, D. Yanagisawa, and K. Nishinari, "A jam-absorption driving system for reducing multiple moving jams by estimating moving jam propagation," *Transportation Research Part C: Emerging Technologies*, vol. 158, p. 104394, 2024.
- C. Liu, F. Zheng, H. X. Liu, and X. Liu, "Optimizing mixed traffic flow: Longitudinal control of connected and automated vehicles to mitigate traffic oscillations," *IEEE Transactions on Intelligent Transportation Systems*, vol. 26, no. 3, pp. 3482–3498, 2025.
- Suzuki, A., Tokumitsu, A., & Nishi, R. (2025). Suppressing secondary shock waves in jam-absorption driving via string-stable support vehicles. *arXiv preprint arXiv:2508.03604*.
- Kreidieh, A. R., Wu, C., & Bayen, A. M. (2018). Dissipating stop-and-go waves in closed and open networks via deep reinforcement learning. *2018 21st International Conference on Intelligent Transportation Systems (ITSC)*, 1475-1480.
- Treiber, M., Hennecke, A., & Helbing, D. (2000). Congested traffic states in empirical observations and microscopic simulations. *Physical review E*, 62(2), 1805.
- Das, A., Kong, W., Sen, R., & Zhou, Y. (2024, July). A decoder-only foundation model for time-series forecasting. In *Forty-first international conference on machine learning*.
- Liu, Y., Zhang, H., Li, C., Huang, X., Wang, J., & Long, M. (2024). Timer: Generative pre-trained transformers are large time series models. *arXiv*

preprint arXiv:2402.02368.

Lundberg, S. M., & Lee, S. I. (2017). A unified approach to interpreting model predictions. *Advances in neural information processing systems*, 30.

Erdmann, J. (2015, March). SUMO's lane-changing model. In *Modeling Mobility with Open Data: 2nd SUMO Conference 2014 Berlin, Germany, May 15-16, 2014* (pp. 105-123). Cham: Springer International Publishing.

Vaswani, Ashish, et al. "Attention is all you need." *Advances in neural information processing systems* 30 (2017).